PRECISION TIMING / ENGINEERING SOFTWARE

MHM2010 / T4
Maser Timing Workbench

Operator and Scientific Manual

FITS and Franks timing, reference 1 PPS figures, MHM2010 DDS/drift and

T4 iMaser 3000 SETFSREL/DRIFTCOMP suggestions, T4 MONIT telemetry and the T4's own monitoring files, stability, further evaluation (gap-aware stability, steering removal, Kalman estimate, three-cornered hat) and native Veusz output against MJD and local ISO time.

Application version	0.3.0 (all content of 0.2.4 to 0.1.0 retained)
Bundled writer	`vszh5_writer.py` 0.0.29 (unchanged from 0.2.4)
Author	William W. Wallace
Date	November 11, 2026
Runtime	Python 3.12 (tested on 3.13.18)
GUI	qtpy / PySide6 / pyqtgraph
License	GPL-3.0-or-later

Review boundary. This application never sends a setting to a maser. The T4 panel can send only the documented read-only queries MONIT, RDFS, GETID and RDREG. Numerical candidates are advisory and require verification of the current settings, sign conventions, reference behavior and application epoch. Historical examples are not settings to enter today. The further-evaluation tabs of release 0.3.0 never change the base fit or the suggested settings.

Deliverable contract: runnable source; live interval fitting; both drift units; FITS and Franks timing inputs; HMD context; stability; further evaluation; Stable32; native `.vszh5` with selectable pages; bookmarked PDF and editable LuaLaTeX; tested examples and a 7z package. Physical instrument certification is outside this release.

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How to read this manual

The manual is ordered by topic: purpose and installation; inputs; the interactive analysis; the reference figures and the suggested settings; the T4; base stability; the further evaluation added in 0.3.0; output; validation; and the appendices.

- **Equations.** Every block of equations is preceded, in its own section, by a table that defines each symbol and its unit. Equation tags name the topic: A (Appendix A, fit, DDS and stability), S (suggested settings), R (reference figures), M, F, D and L (T4 telemetry, SETFSREL, DRIFTCOMP and monitoring files), C, T and V (time display, exact time text and Veusz time axes), G, H, N and X (gap-aware stability), W (steering), K (Kalman) and Q (three-cornered hat).
- **Appendix B** lists every symbol of the manual with its unit and the equations that use it. A symbol used with two meanings in different topics is listed once per topic.
- **Appendix C** describes every stability trace: in the GUI, on the Veusz pages, as datasets and as file columns.
- **Numbers.** Results quoted from exports carry every digit the export holds (the shortest decimal that reads back to the same binary64 value).

Purpose and verified instrument relationship

The instrument mapping follows the MHM2010 manufacturer's guide [1]; stability definitions follow NIST SP 1065 [2]; and interchange files follow the Stable32 manual [3]. These references establish the analytical framework, not a certification of the installed system.

The workbench estimates residual maser-to-reference fractional frequency and frequency drift from measured timing offsets. Housekeeping telemetry identifies instrument context and changes; it is not an independent frequency reference. The Green Bank system logs switched one-pulse-per-second comparisons involving Site1Hz, MicroSem and GPS channels [4].

The MHM2010 manual gives the exact relationship between maser oscillation frequency, the external synthesizer number and the standard output frequency. An increase in the synthesizer number lowers output frequency; the nominal calibration is 7.04024×10^{-17} per least-significant displayed digit [1]. The application derives its local gain from the exact denominator, rather than treating the nominal calibration as exact.

The supplied screenshot identifies the two target controls as **External DDS Setting** and **Drift Correction**. The manufacturer's manual explicitly offers **DDS Units/day** and **pp** 10^{18}**/day** in the drift control [1]. Both are computed simultaneously.

Table 1. MHM2010 items documented in the manufacturer's guide and their treatment

Documented item	Treatment in this release
DDS versus output direction	Increasing DDS decreases output frequency [1].
DDS precision	One displayed last digit is 10^{-7} in the synthesizer number [1].
Drift-unit choices	Both DDS units/day and parts in 10^{18}/day are available [1].
External control	Front-panel synthesizer selection must permit external control [1].
Drift-field sign	Not explicitly stated in inspected Rev. J; user selects and verifies it.
Numeric limits and cadence	Not specified in inspected Rev. J; not invented by this program.
Housekeeping selector codes	Retained as raw logged values; not decoded into an enable assertion.

Review boundary. A phase offset cannot be removed merely by selecting a zero steady frequency error. This software does not plan phase slews or write the instrument. It estimates slope and curvature and leaves the constant phase offset unchanged.

Release structure

Table 2. Package structure of release 0.3.0

Part	Content
`MHM2010_Correction_App.py` 0.3.0	GUI entry point (tabs, menus, OUTPUT checkboxes)
`maser_io`, `maser_core`, `maser_plots`	inputs, fit and settings science, reference figures
`maser_eval.py` 0.0.1 (new)	further evaluation: gap-aware stability, steering removal, Kalman, three-cornered hat
`gui_evaluation.py` 0.0.1 (new)	the four "Eval:" tabs
`maser_export.py` 0.3.0	export bundle, Veusz pages, `ExportContent` page choices, evaluation files
`analyze_maser.py` 0.3.0	batch path with page and evaluation flags
`t4_*.py`, `tools/T4_Reboot.py`	T4 telemetry, monitoring files, emulator; separate reboot script
`vszh5_writer.py` 0.0.29	native Veusz HDF5 writer (unchanged)
`docs/`	this PDF, its LuaLaTeX source, validation report, writer notes
`examples/`, `validation/`, `tests/`	historical results, renders and GUI captures, 174 tests

What is new in release 0.3.0

- **Further evaluation in four new tabs** (section "Further evaluation (release 0.3.0)"): *Eval: gap-aware stability* (OADEV, MDEV, TDEV, OHDEV on a regular grid with gap markers, lag-1 noise identification and confidence intervals, NIST SP 1065 [2]), *Eval: steering removal* (the MHM2010 steering recorded in the HMD integrated to phase and removed), *Eval: Kalman estimate* (phase, frequency and drift with a three-state clock model) and *Eval: three-cornered hat* (individual Allan deviations of three clocks). A new **Evaluate** menu calculates each tab.
- **The base behavior is unchanged.** The fit, the Results box, the suggested settings, the reference figures and the base Stability tab are computed exactly as in 0.2.4; nothing in the evaluation tabs is written back (tested: identical suggestion JSON before and after).
- **Veusz pages to generate.** OUTPUT has checkboxes for seven base page groups (all on by default, giving the 0.2.4 file) and four evaluation page groups (off by default), with *Defaults* and *All* buttons (section "Veusz pages to generate"). Telemetry plots can be left out, and steering-removed pages added. The choice is saved with the settings.
- **Batch flags.** `--no-reference-figures`, `--no-timing-pages`, `--no-overview`, `--no-stability-pages`, `--no-hmd-plots`, `--no-t4-plots`, `--no-t4-monitoring-plots` and the evaluation flags `--gap-stability`, `--remove-steering`, `--kalman`, `--tch` with their options.
- **New files.** `evaluation_gap_stability.csv`, `evaluation_steering_removed.csv`, `evaluation_kalman.csv`, `evaluation_three_cornered_hat.csv`; `analysis.json` adds `export_content`, `evaluation` and `evaluation_files`; `review.md` adds "Veusz content" and "Further evaluation". Every number is exact.
- **Fix.** Switching to another dataset after a fit could leave the old fit in place and raise an `IndexError` traceback (0.2.4); the fit, candidate, suggestion and stability result are now cleared when the channel changes.
- **Documentation.** This manual is reorganized by topic, with a variable table before every equation block, Appendix B (all variables), Appendix C (all stability traces), and lists of tables and figures in the contents and the bookmarks.
- **Tests.** 174 pass (159 retained, 15 new); the default export is identical to 0.2.4 (dataset hashes, tags, pages and script).

Installation and first run

Use Python 3.12 in an isolated environment. No fixed personal directory names are embedded in the application. Run the following inside the extracted package:

```
py -3.12 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe MHM2010_Correction_App.py
```

With mamba, the equivalent is:

```
mamba create -n maserwb python=3.12 numpy scipy pandas astropy h5py qtpy pyside6 pyqtgraph
mamba activate maserwb
python MHM2010_Correction_App.py
```

The supplied Windows launcher uses the adjacent environment if present, otherwise Python's 3.12 launcher. Source files have author/revision headers, semantic versions, Spyder cell markers and explanatory comments. In Spyder, select the same environment; a dedicated console is preferable if a GUI loop is already active.

The default backend is PySide6 through qtpy. The application does not import PySide6 directly. pyqtgraph is configured to match the selected Qt backend. PyQt6 may be selected explicitly through `QT_API`, but that alternate backend was not validated.

Dependencies and offline behavior

The runtime requirements are NumPy, SciPy, pandas, Astropy, h5py, QtPy, PySide6 and pyqtgraph; the evaluation module uses SciPy's `stats` (chi-squared and normal quantiles) and `optimize` (non-negative least squares). The supplied environment-version list records the packages used for testing. Installing dependencies requires package access; the scientific analysis itself requires no web service, API key, instrument socket or serial connection.

Veusz is optional for viewing or rendering output, not for generating `.vszh5`. Test dependencies include pytest and AllanTools [25]. The included manual was compiled with LuaLaTeX; run the command three times from the docs folder (contents, lists of tables and figures, and references settle on the third run):

```
lualatex -interaction=nonstopmode -halt-on-error MHM2010_T4_Workbench_Manual_v0.3.0.tex
```

Quick workflow

1. Select raw OnePpsDeltas FITS/FITS.GZ files or Franks files using the GUI or drag-and-drop.
2. Load HMD housekeeping separately; select the intended MicroSem/GPS pair.
3. Choose a continuous acquisition run and a clean fit interval.
4. Compare linear and quadratic behavior and inspect residuals and other references.
5. Enter the DDS and enabled drift at the intended application epoch.
6. Verify signs and operating state; inspect both drift-unit representations.
7. Compute stability with an explicit sampling tolerance and detrending choice.
8. Optionally run the evaluation tabs (Evaluate menu) to see the gap-aware stability with confidence intervals, the maser without its recorded steering, the Kalman estimate and a three-cornered hat; they inform the review but do not change the suggestion.
9. Tick the Veusz pages to generate and export to a new results folder; retain the source hashes and review record.

Review boundary. This release was tested in a Linux cloud environment. Actual Windows file dialogs, display scaling, Stable32 application import and physical maser operation remain untested.

Input data and provenance

Preferred input: raw OnePpsDeltas FITS

The parser requires the `OnePpsDeltas` binary table with `DMJD`, `CHANNELA`, `CHANNELB` and `DELTAT`. It verifies UTC time metadata and declared time/phase units. Files marked simulated are rejected. Blank channel names, nonpositive timestamps and padding rows are rejected. FITS `TNULL` values are honored when supplied.

Every measured channel pair stays separate. The direct `MicroSem -> CNS_GPS1` pair is preferred over subtracting sequential Site1Hz observations. Reverse-direction pairs are not merged into the direct pair: their measurements occur at different times and can include fixed channel-path asymmetries. Valid raw zero values and -9999 ns observations are not automatically treated as missing.

Rows are sorted by timestamp. Identical duplicates are removed and counted; different values at one identical pair timestamp cause a failure rather than an arbitrary choice. The parser records input basenames, sizes and SHA256 hashes, without exporting personal absolute directory paths.

Alternative input: Franks t00new

Franks files are reduced timing data. The parser locates the named MJD header, requires the nanosecond declaration and preserves each available timing column. Values -9999 and the historical -999 sentinel are removed in this adapter only.

Hourly median timestamps are often irregular. They can be fitted using actual timestamps, but cannot automatically be treated as a uniform stability record (the gap-aware evaluation places them on a sequential grid instead, section "Eval: gap-aware stability"). The original reduction windows and raw samples cannot be recovered from the reduced values.

FITS and Franks inputs together

- FITS series are raw channel pairs keyed "A -> B"; Franks series are hourly medians keyed by column (e.g. `MSM-GPS`). Keys cannot collide; nothing is merged or interpolated.
- **Month boundaries.** Consecutive `YYYY_MM.t00new` files repeat the first hours of the next month. Identical repeats are dropped. A disagreeing repeat (found in the supplied files at MJD 61222.01917, Site1Hz 0 vs 1) keeps the row from the file named for that row's month, and the conflict is listed in the audit tab.
- **Gaps.** The 0.1.0 rule refused a fit across an acquisition gap. "Allow fit across acquisition gaps" (default on for Franks data) fits through gaps and reports each one. The supplied 2026 MSM-GPS series has nine gaps of 5 to 99 h; inspect the residuals for a phase step.

HMD telemetry

HMD is parsed as a timestamped comma-delimited table. It must include MJD and the external synthesizer column. All numeric housekeeping channels remain available for display. Large discrete DDS changes and selector-code changes are marked as candidate events; gradual drift ramps are not called steps.

The event detector uses a configurable function argument with a GUI default of 50 least-significant DDS digits. This is an application heuristic, not a manufacturer alarm threshold. Alarm/status channel voltages remain raw values. Missing HMD coverage is reported; historical telemetry is never silently used as the current state at an unrelated epoch. In 0.3.0 the HMD DDS record also drives the steering-removal evaluation (section "Eval: steering removal").

Timing overview and the supplied Veusz exemplar

The overview creates UTC-hour medians for available Site1Hz comparisons, direct MicroSem/GPS measurements and a separately labeled difference of Site1Hz medians. All raw measured pairs remain in the export. Missing channels such as RA or TAC are not fabricated.

Each available overview channel has MJD and proper UTC-axis export pages. This reproduces the useful dataset families in the supplied Franks Veusz example without copying its all-missing channels or invalid legacy dates. UTC-hour binning is not claimed identical to the historical reduction windows.

Interactive analysis

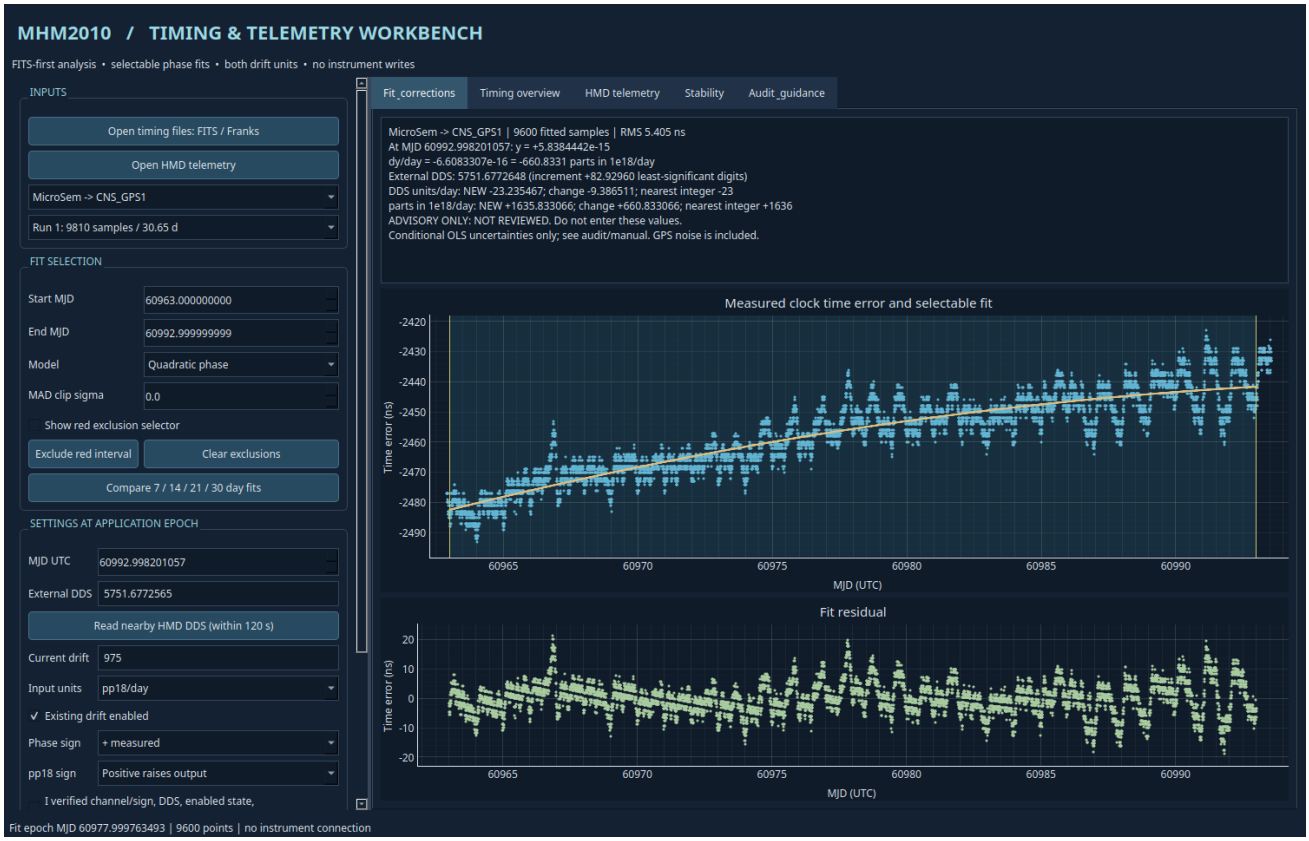


Figure 1. GUI tab "Fit and corrections": fit interval (blue), exclusion selector (red), phase and residual plots.

Fit and corrections tab. The blue region selects the fitting interval. The phase and residual plots share their horizontal view. Numeric MJD limits provide precise selection without requiring a mouse.

The red region is an independent exclusion selector. Add its interval to the exclusion list or clear the list. Excluded observations remain in the input and export, with an explicit used-in-fit mask. Optional iterative MAD residual clipping is disabled by default. It can reject outliers, but it changes noise statistics and must not conceal physical events.

A linear phase model estimates frequency only; the drift field is not estimated and should not be changed on that basis. A quadratic phase model estimates frequency and frequency drift. Internal scaling improves numerical conditioning without changing the physical units of the returned coefficients.

The 7/14/21/30/90-day comparison button reports trailing-window fits in the audit tab. These are unclipped quadratic diagnostic fits, with actual span and sample count shown. They are not confidence intervals.

Table 3. GUI tabs of release 0.3.0, in order

Tab	Content
Fit and corrections	fit interval, exclusions, model, Results box with the suggested changes
1 PPS reference plots	the operator's two figures (deltas with statistics; data and fit over residuals)
Timing overview	every channel (hourly medians for FITS)
HMD telemetry	MHM2010 housekeeping columns and DDS events
T4 telemetry / T4 monitoring files	T4 MONIT and the T4's own logs
Stability	base OADEV, MDEV, OHDEV and TDEV (uniform record; section "Stability analysis and Stable32 (base tab)")
Eval: gap-aware stability	0.3.0, section "Eval: gap-aware stability"
Eval: steering removal	0.3.0, section "Eval: steering removal"
Eval: Kalman estimate	0.3.0, section "Eval: Kalman estimate"
Eval: three-cornered hat	0.3.0, section "Eval: three-cornered hat"
Audit & guidance	audit trail, window comparison, export log

State and stale-result handling

Changing the fit interval/model invalidates old stability results. In 0.3.0 the evaluation results are invalidated too, by any change of the channel, fit, interval, application epoch or phase sign; an export recalculates every ticked evaluation from the current fit. Changing the channel clears the fit, candidate, suggestion and stability result (the 0.2.4 stale-fit traceback is fixed). Settings can be saved without source-directory paths, then restored after selecting the original files; they include the Veusz page choices and the evaluation options. Restoring settings clears operator acknowledgments. Export creates a new folder; it never silently replaces an earlier result directory.

Review boundary. The plots are interactive analysis of selected files, not streaming acquisition. Large file reads run outside the GUI thread; fitting, overview construction, evaluation and export may briefly occupy the GUI thread for very large datasets. There is no serial or network maser interface in this release.

Time display: MJD or ISO 8601 calendar time

Choose **View** → **Time display (axes, dates, ranges)**:

Table 4. Time display choices

Menu entry	Example	Shortcut
MJD (UTC)	`61349.86612`	Ctrl+Shift+M
Calendar date-time, local (ISO 8601)	`2026-11-05T15:47:13-05:00`	Ctrl+Shift+L
Calendar date-time, UTC (ISO 8601)	`2026-11-05T20:47:13+00:00`	Ctrl+Shift+U

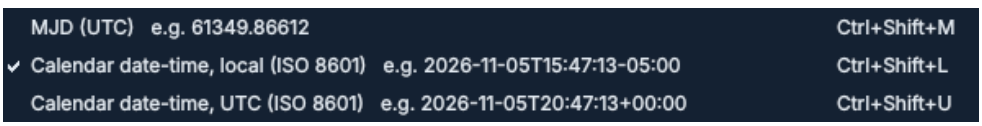


Figure 2. View menu: Time display (axes, dates, ranges).

The choice applies at once to: the fit, residual, HMD, overview, reference-figure, T4 telemetry, T4 monitoring-file and evaluation time axes; the Start, End and application-epoch fields (they show and accept the chosen form); the run list; the result text and status bar (calendar times keep the MJD in brackets for the audit trail); and audit messages. It is remembered in `workbench_view.ini` beside the program and saved with the analysis settings. The data, fits and exported numbers are always MJD

(UTC); only the display changes. The `.vszh5` always contains every time page against both MJD and local ISO time.

Typing a time. In calendar modes a field accepts `2026-11-05T15:47:13-05:00`, the one-digit offset `-5:00`, `Z`, a date alone, a time without offset (read as local in local mode, UTC in UTC mode) or a plain MJD. Up/down steps one hour (one day in MJD mode). The batch script's `--start`/`--end` accept the same forms (no offset = UTC).

Conversion and ticks. The display converts as follows.

Table 5. Variables of the time-display conversion (C1)

Symbol	Meaning	Unit
MJD	Modified Julian Date of the sample, UTC	day
t^{Unix}	seconds since 1970-01-01T00:00:00 UTC (POSIX, no leap seconds)	s
t^{local}	local wall-clock seconds on the same scale	s
$o(t)$	UTC offset of the processing computer at instant t, daylight saving included	s
40587	MJD of 1970-01-01	day

$$t^{\mathrm{Unix}} = (\mathrm{MJD} - 40587) \times 86400\ \mathrm{s}, \qquad t^{\mathrm{local}} = t^{\mathrm{Unix}} + o(t^{\mathrm{Unix}}), \tag{C1}$$

so a July time at Green Bank shows `-04:00` and a November time `-05:00`. Ticks fall on round calendar times (pyqtgraph's date logic, daylight-saving aware) and are labelled with the full ISO time on two lines (date, then `Thh:mm:ss±hh:mm`); ticks closer than one label width are dropped so labels never overlap. The axis title gives the offsets in the range, e.g. "Local date-time, ISO 8601 (UTC−05:00 then −04:00)". "Julian" in common speech often means this calendar view; strictly, MJD is the Julian-day count (JD−2 400 000.5) and the calendar dates are Gregorian.

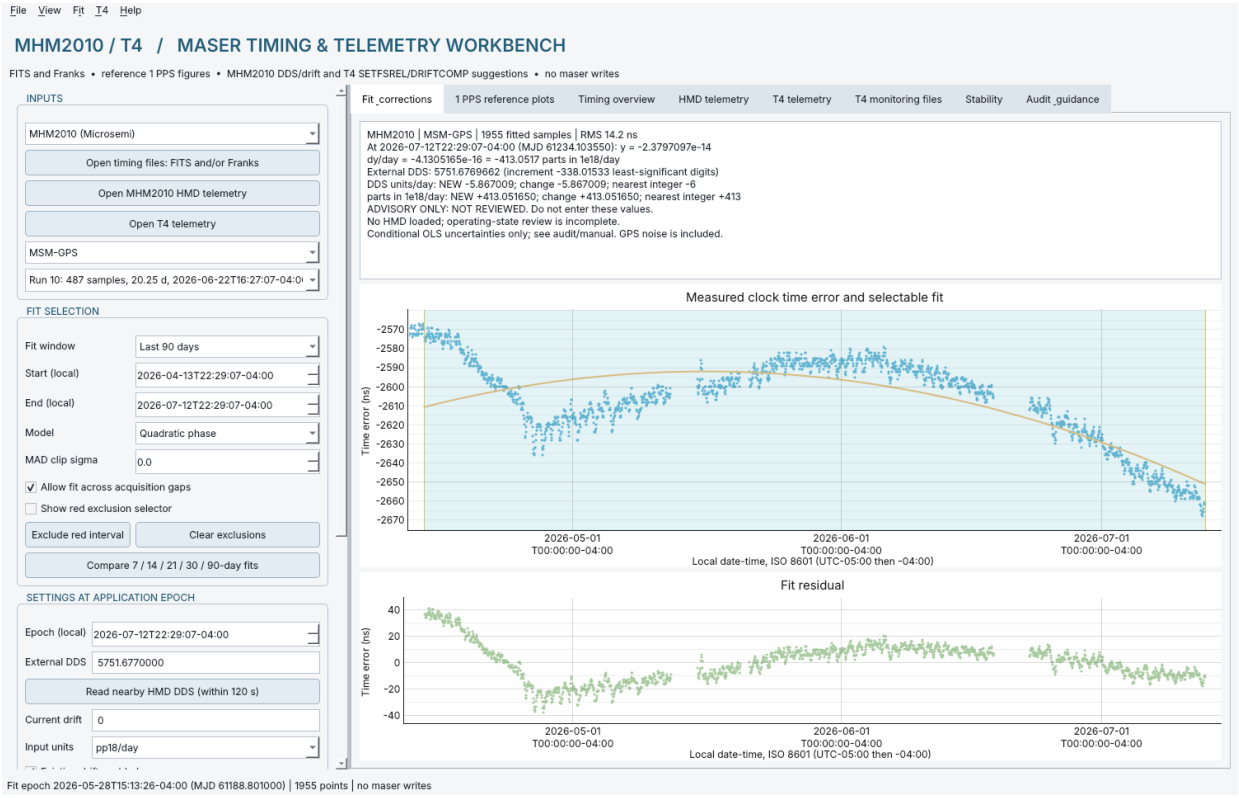


Figure 3. Local ISO 8601 display, last 90 days of the 2026 Franks MSM-GPS channel (Light theme).

Fit windows: last 7 to 90 days

FIT SELECTION → **Fit window** (or the **Fit** menu) sets the blue interval to the last 7, 14, 30 or 90 days ending at the last sample, or to all data, and moves the application epoch to the last sample. A preset is enabled only when the loaded channel spans it (its tooltip gives the span otherwise). If the window contains acquisition gaps, "Allow fit across acquisition gaps" is switched on and logged; check the plot for phase steps. Dragging the blue interval returns to "Custom". **Compare 7/14/21/30/90-day fits** (or Fit → Compare window lengths) fits each trailing window ending at the selected end, skipping windows longer than the data; results go to the audit tab. In batch, `--last-days 30` or `90` does the same and stops if the data are shorter.

On the 2026 Franks MSM-GPS data (193 days), the 90-day quadratic window ending 2026-07-13 02:29 UTC gives $y = -2.38 \times 10^{-14}$ and drift -413 parts in 10^{18}/day with 14.2 ns RMS; the 30-day window gives $y = -3.71 \times 10^{-14}$, -686 parts in 10^{18}/day and 3.6 ns RMS. The difference is the window sensitivity this comparison exists to expose; the 90-day window spans gaps and a visible phase excursion in April.

Reference 1 PPS figures

The operator's two figures (the deltas with their statistics; the data and fit over the residuals) are drawn in the GUI tab "1 PPS reference plots" and are the first pages of every `analysis.vszh5`. They keep the operator's notation, so the symbols below are local to this section (Appendix B lists them under "Reference figures").

Definitions

For the samples of a figure, the horizontal coordinate is days since the first sample shown and the vertical coordinate is the 1 PPS time difference. The deltas figure uses every sample of the selected interval; the fit figure uses the samples actually fitted (operator exclusions removed, drawn hollow and grey).

Table 6. Variables of the reference 1 PPS figures (R1–R5)

Symbol	Meaning	Unit
N_s	number of samples in the figure	1
MJD_i, MJD_0	time of sample i; first sample shown	day
x_i	$\mathrm{MJD}_i - \mathrm{MJD}_0$, horizontal coordinate	day
u_i	1 PPS time difference of sample i (the figure's "y")	ns
$\bar{u}$, std	mean and population standard deviation of u_i	ns
$\bar{x}$	mean of x_i	day
S_{xx}, S_{xy}	$\sum (x_i - \bar{x})^2$; $\sum (x_i - \bar{x})(u_i - \bar{u})$	day^2; ns day
a_1, a_0	slope and intercept of the least-squares line $u = a_1 x + a_0$ (printed "$y = a_1 x + a_0$")	ns/day; ns
r_i	residual $u_i - a_1 x_i - a_0$	ns
σ^2	variance of independent residuals (model)	ns^2
s	residual standard deviation estimate, $s^2 = \sum r_i^2/(N_s - 2)$	ns
$s(a_1)$, $s(a_0)$	standard errors of a_1 and a_0	ns/day; ns
y_L	fractional frequency of the straight line (the figure's "rate")	1
ϵ	phase sign, ± 1 (ϵu = maser minus reference)	1
N	MHM2010 DDS setting (external synthesizer number)	DDS units
a, q	2×10^{-7} (DDS coefficient); 10^{-7} (one count)	1; DDS units
Δn	synthesizer adjustment of the figure	counts

The boxes report the mean and the population standard deviation (as `numpy.std`):

$$\bar{u} = \frac{1}{N_s}\sum_i u_i, \qquad \mathrm{std} = \sqrt{\frac{1}{N_s}\sum_i (u_i - \bar{u})^2}. \tag{R1}$$

Straight-line fit and its standard errors

The symbols are defined in Table 6. The least-squares line $u = a_1 x + a_0$ minimizes $\sum r_i^2$. Setting $\partial/\partial a_0 = \partial/\partial a_1 = 0$ gives the normal equations, whose solution is

$$a_1 = \frac{S_{xy}}{S_{xx}}, \qquad a_0 = \bar{u} - a_1 \bar{x}. \tag{R2}$$

For independent residuals of variance σ^2, $\mathrm{Var}(a_1) = \sigma^2/S_{xx}$; because $\mathrm{Cov}(\bar{u}, a_1) = 0$, $\mathrm{Var}(a_0) = \mathrm{Var}(\bar{u}) + \bar{x}^2\,\mathrm{Var}(a_1) = \sigma^2\left(\frac{1}{N_s} + \frac{\bar{x}^2}{S_{xx}}\right)$. With s^2 in place of σ^2:

$$s(a_1) = \frac{s}{\sqrt{S_{xx}}}, \qquad s(a_0) = s\sqrt{\frac{1}{N_s} + \frac{\bar{x}^2}{S_{xx}}}, \qquad y_L = \frac{a_1\,[\mathrm{ns/day}] \times 10^{-9}}{86400\ \mathrm{s}}. \tag{R3}$$

These are conditional IID errors; colored noise and GPS systematics are not included [2]. Check against the operator's figure: $N_s = 741$, $s = 3.49$ ns over 31 days gives $S_{xx} \approx 741 \times 31^2/12$, $s(a_1) \approx 0.014$ ns/day

and $s(a_0) \approx 0.257$ ns, as printed. The gap-aware stability and the Kalman estimate of 0.3.0 show how much larger the uncertainty is when the identified noise is not white (section "Results on the supplied data").

Synthesizer adjustment line

The symbols are defined in Table 6. **MHM2010.** Appendix A gives the fractional change per DDS unit, $\partial y/\partial N = -a/(284.08 + aN)$ [1]. One count is $q = 10^{-7}$ DDS units, so cancelling $y = \epsilon y_L$ requires

$$\Delta n = \frac{-\epsilon y_L}{-aq/(284.08 + aN)} = \frac{\epsilon y_L\,(284.08 + aN)}{2 \times 10^{-14}}. \tag{R4}$$

For $y_L = -3.15 \times 10^{-14}$ this is -447 counts, the value in the operator's figure; the full DDS calculation of Appendix A agrees to 10^{-9} relative (test suite).

T4. The line shows the SETFSREL entry and the synthesizer change of section "Frequency setting: SETFSREL" for the same straight-line rate.

Review boundary. The figure's adjustment uses the straight-line mean rate over the plotted span, as the operator's figures do. The settings panel evaluates the selected model (linear or quadratic) at the application epoch. With a quadratic fit the two can differ; enter only the reviewed settings-panel values.

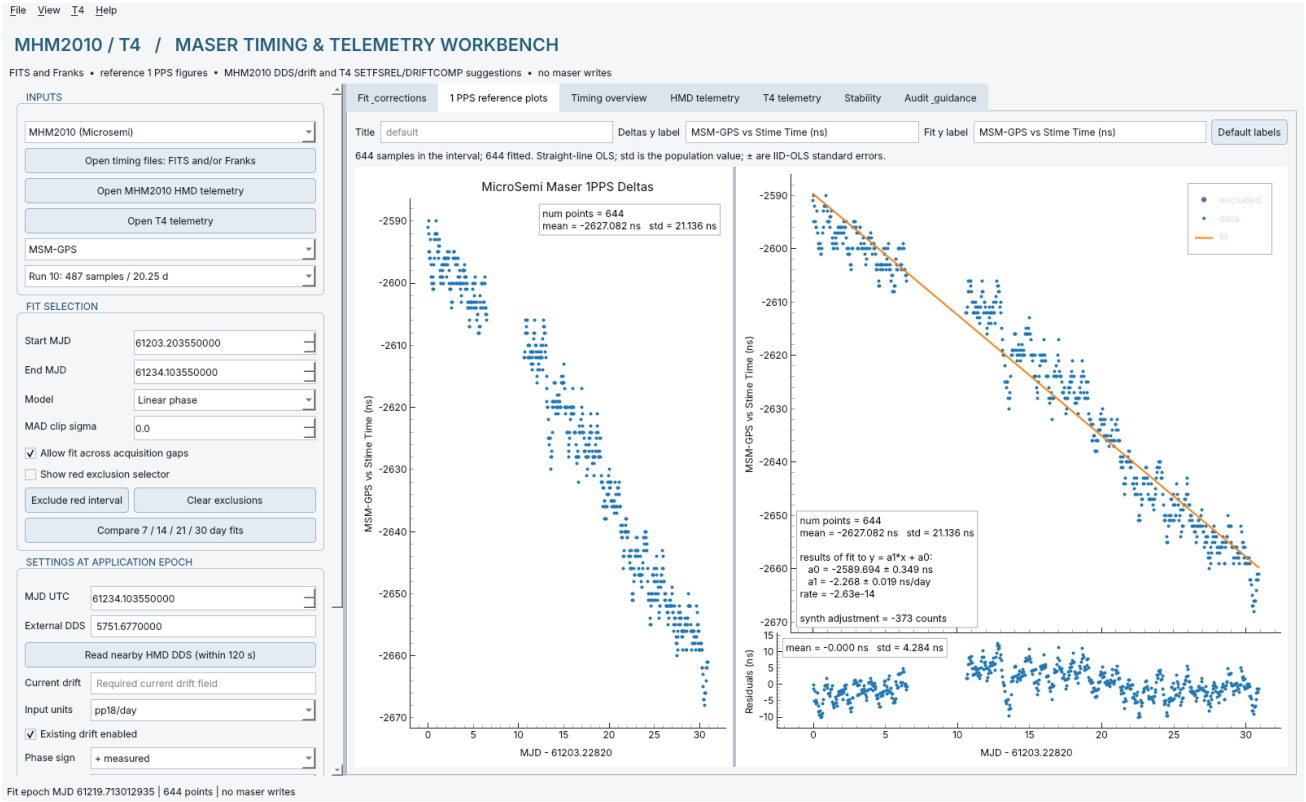


Figure 4. GUI tab "1 PPS reference plots" (Light theme) on the operator's Franks files, June–July 2026, MSM-GPS.

Correction interpretation and manual review

One epoch, two absolute settings

The phase fit is centered on an internal reference epoch. Frequency error is evaluated at the separate intended application epoch. The old DDS must be the setting at that application epoch, not the setting at the start of the data or at the centered fit epoch.

The new external DDS is an **absolute setting**, not a number to add again. Its displayed increment is also reported in least-significant DDS digits. The new drift is the existing enabled drift plus a residual correction, with the sign and units handled explicitly. An existing disabled drift contributes zero regardless of a stale numeric value in its field.

The pp18/day input field and DDS-units/day input field are alternative representations of the same ramp. The program reports both new values and both increments. Do not apply both increments sequentially.

Sign and precision

For the adopted timing definition, measured arrival(B) minus arrival(A) equals clock(A) minus clock(B). A direct MicroSem-to-GPS pair therefore uses the positive sign; a reverse pair uses the negative sign. This channel interpretation must be checked against the installed measurement wiring and logging convention.

Positive physical DDS units/day means the DDS number increases with time, which lowers output frequency [1]. The installed pp18 sign is separately selectable: positive raises output frequency, or positive raises DDS. The default is an explicit assumption, not a claim that Rev. J documents that sign.

The external DDS is rounded to seven decimal places. Both drift values are also shown without integer rounding and with nearest-integer alternatives. The integer alternatives are illustrative: verify what entry precision the installed masermon accepts. pp18 integer increments and DDS integer increments do not have the same effective frequency-drift resolution.

Before entering any candidate

- **Reference**: confirm the GPS channel is healthy, not stepped or being steered unexpectedly.
- **Instrument**: confirm external synthesizer selection, current DDS, drift enable state and units.
- **Sign**: verify the instrument's installed pp18 convention; do not test it by an uncontrolled adjustment.
- **Time**: use a current, defensible application epoch and known setting at that epoch.
- **Fit**: inspect residuals, HMD changes, outages and window sensitivity.
- **Evaluation (0.3.0)**: compare the base drift with the steering-removed drift and the Kalman uncertainty; a drift change smaller than its Kalman 1σ is not supported by the data.
- **Record**: preserve old settings, intended values, operator review and subsequent comparison data.

No checkbox turns this software into a certified controller. Values remain advisory even after the operator review fields are acknowledged.

Suggested settings: frequency and drift

Equations

The suggestion is made from the selected phase model and the channel sign at the application epoch, with the DDS relations of Appendix A, equations (A13)–(A19). The symbols are those of Appendix A.

Table 7. Variables of the suggested settings (S1–S4)

Symbol	Meaning	Unit
c_0, c_1, c_2	phase-model coefficients, $x = c_0 + c_1 d + c_2 d^2$	ns; ns/day; ns/day^2
d, d_a	elapsed days (SI) from the fit epoch; at the application epoch	day
ϵ	phase sign, ± 1 (ϵx = maser minus reference)	1
y	residual fractional frequency of the maser at the application epoch	1
r	residual fractional-frequency drift	day^{-1}
N, N_{new}	current and new DDS setting	DDS units
a	DDS coefficient 2×10^{-7} [1]	1
D	synthesizer denominator $284.08 + aN$ (nominal 284.08 when N is not entered)	1
k_N	a/D, fractional output change per DDS unit (magnitude)	DDS units^{-1}
q	one DDS count, 10^{-7}	DDS units
η	pp18 sign convention: $+1$ positive raises output, -1 positive raises DDS	1
Δn	DDS change	counts
ΔP	drift-field change in pp18/day	10^{-18}/day
Δv	drift-field change in DDS counts/day	counts/day
v_{old}	current drift field in DDS counts/day	counts/day
round_q	rounding to a multiple of q	–

$$y = \epsilon\,\frac{(c_1 + 2c_2 d_a)\,10^{-9}}{86400}, \qquad r = \epsilon\,\frac{2c_2\,10^{-9}}{86400}\ \text{per day}, \tag{S1}$$

$$\Delta n = \frac{D}{a}\,\frac{y}{q} \quad (\text{DDS change, counts}), \tag{S2}$$

$$\Delta P = -\eta\,\frac{r}{1+y}\,10^{18} \quad (\text{drift change, pp18/day}), \tag{S3}$$

$$\Delta v = \frac{r}{k_N q}\ \big(+\,v_{\text{old}}\,y \text{ when the current field is in DDS counts/day}\big) \quad (\text{drift change, DDS counts/day}). \tag{S4}$$

Without a DDS setting the nominal $D = 284.08$ is used (a relative change of 4×10^{-6} at $N = 5752$). With the current values: $N_{\text{new}} = \text{round}_q(N + q\Delta n)$ and drift field = current + change. T4: `SETFSREL` from Eq. 12 of [8] (39-bit step, equation (F4)) and $\texttt{DRIFTCOMP}_{\text{new}} = \texttt{DRIFTCOMP}_{\text{now}} - r$ [9] (equation (D3), with r already carrying ϵ). A linear model estimates no drift; the box then says so. The arithmetic is the same as `correction()` and `t4_settings()` (tested to 10^{-13} or identical). In 0.2.4 the same equations were written with a_0, a_1, a_2, τ and D for the coefficients, the days and the drift; the symbols are now those of Appendix A, and the arithmetic is unchanged.

Where the two changes are stated

Table 8. Where the suggested frequency and drift changes appear

Place	Content
Fit & corrections, Results	"SUGGESTED CHANGES": the box text plus the exact values and notes; below, the new absolute settings when the current DDS and drift are entered
1 PPS reference plots, fit figure	box "suggested ... changes" at the upper left; the y axis leaves room above the data so the box never covers points
`analysis.vszh5`	the same box on `Ref1PPS_<channel>_fit_MJD` and `_fit_local`; datasets `Suggested_<maser>_*` (one value each, exact), `_full_precision_text`, `_box_text`
`analysis.json`, `review.md`	`suggested_settings`; exact-value table
`analyze_maser.py`	printed `suggested_settings`
Eval: Kalman estimate (0.3.0)	settings from the Kalman estimate, marked EVALUATION ONLY; never in the box, the Results or `suggested_settings`

Example: last 30 days of the 2026 Franks MSM-GPS data

- Quadratic phase, 622 points; epoch `2026-07-12T22:29:06.72-04:00` (MJD 61234.10355).
- $y = -3.708732994065008 \times 10^{-14}$; $r = -6.864816602325095 \times 10^{-16}$/day.
- **DDS change** -526.79 ± 10.49 **counts; drift change** $+686.48 \pm 47.16$ **pp18/day** $= -9.7508$ **DDS counts/day** (positive pp18 raises the output).
- With DDS 5751.6770000 and drift -120 pp18/day: new DDS 5751.6769473 and new drift field $+566.48$ pp18/day.

The straight-line "synth adjustment" of the same figure is -379.26 counts (interval-mean rate); the quadratic gives -392.00 counts at the mean sample time (MJD 61220.28; uneven sampling around the 5-day gap) and -526.79 at the end, $13.82\,\mathrm{d} \times 9.75$ counts/day later. The $\pm$ values are conditional OLS uncertainties; GPS and colored noise are not included.

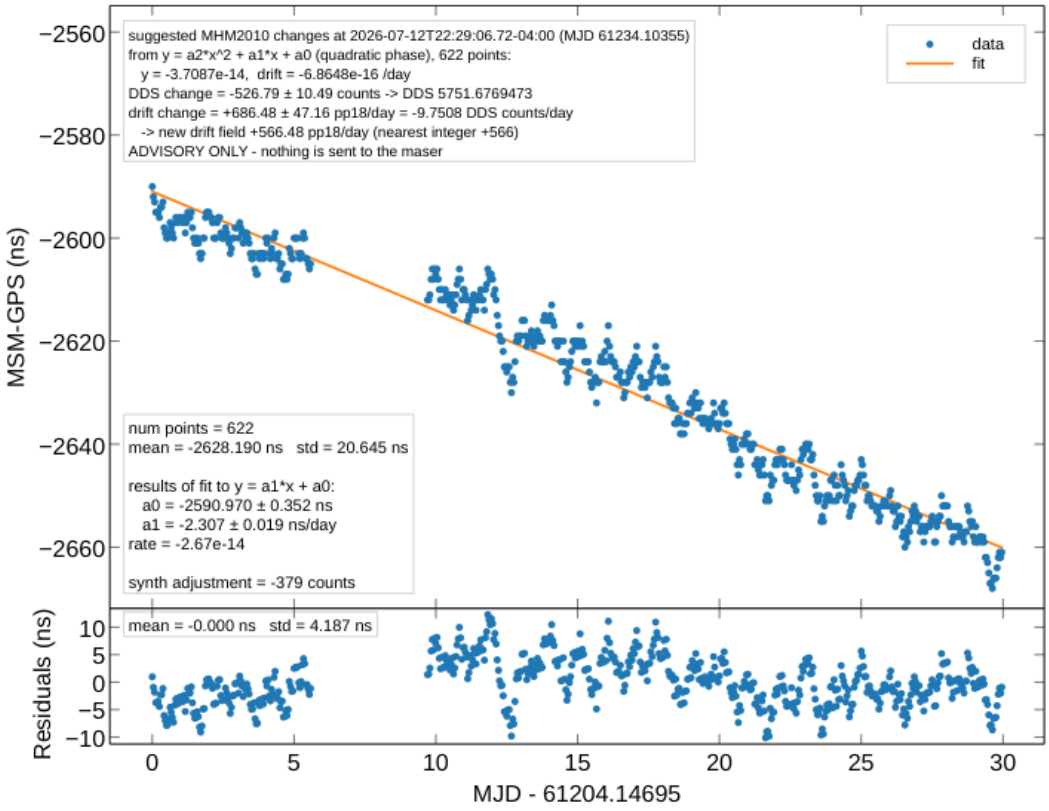


Figure 5. Veusz 3.6.2 render of `Ref1PPS_MSM_GPS_fit_MJD`: the figure's results box (lower left) and the suggested-changes box (upper left).

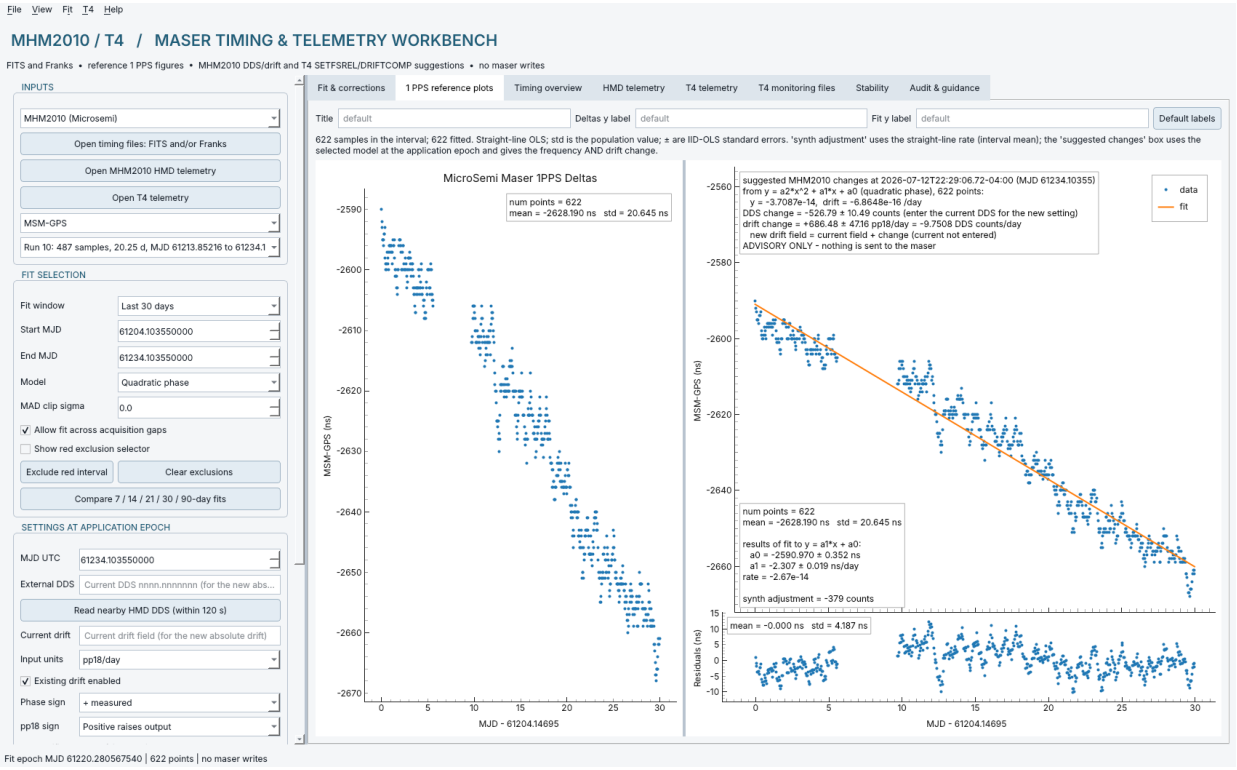


Figure 6. The GUI with empty DDS and drift fields: both changes are stated.

Review boundary. Enter only reviewed values. Confirm the pp18 sign convention of the installed masermon, the channel sign, the epoch and the maser state; the HMD shows whether a drift is already acting on the DDS.

T4 iMaser 3000

MONIT telemetry

The reply to `MONIT;` is `$` + 113 hexadecimal characters + CR LF (Type 1, Card FPGA T4S-724) or CR (Type 2, IO card LAN T4S-711) [8]. The channels are 32 of 12 bits, 8 of 8 bits and the main-PLL lock bit.

Table 9. Variables of the MONIT decoding (M1)

Symbol	Meaning	Unit
L_{hex}	number of hexadecimal characters after `$`	characters
i	channel number, 1 to 41	1
field_i	hexadecimal field of channel i (3 or 2 characters, or the lock bit)	–
c_i	unsigned integer code of channel i	code
g_i	gain of channel i, Table 6 of [8] (negative for the -15 V rail)	V/code (or unit/code)
v_i	decoded value of channel i	V (or channel unit)

$$L_{\text{hex}} = 32 \cdot 3 + 8 \cdot 2 + 1 = 113, \qquad v_i = g_i\, c_i, \quad c_i = \texttt{int}(\text{field}_i, 16). \tag{M1}$$

Gains g_i are Table 6 of [8], identical to the operator's repository table [10]; the -15 V rail has a negative gain. Channels 21, 37 and 39 are "Not used". Channel 40 (LOCK 100 MHz) is an 8-bit analog voltage; channel 41 is the lock bit (1 locked, 0 unlocked; any other code is flagged). Decoding the manual's example reply gives 4.765 V for channel 40; the operator's web-page screenshot shows 4.764 V.

Inputs. The recorder's `t4_raw_frames.csv` (re-decoded from the raw replies), its `t4_telemetry.csv` (raw codes re-scaled), or text logs with one `<time> $<frame>` per line. Times must be MJD or ISO 8601 with an offset.

Live polling (T4 tab) sends only `MONIT;`, `RDFS;`, `GETID;` and `RDREG=…;` over UDP port 14000 [9]. Each attempt uses a fresh socket with bounded retries; datagrams from other peers are ignored. Since 0.2.1 the local UDP port defaults to 14000 for a real T4: the vendor example and `T4_Maser_Comms.py` bind it ("seems one must bind the port to local port as well" [10]), so the T4 probably answers to port 14000 rather than to the sender's port; the manual does not say. The emulator uses an ephemeral port. Raw replies are written before decoding, into a folder stamped once at creation (`T4_telemetry_YYYYMMDDTHHMMSS±hhmm`). The preview buffer is bounded and can be flushed or offloaded to a pickle. A non-loopback address requires the approval box. "Use loopback emulator" serves the manual's example reply. Since 0.2.4 the recorder CSVs write MJD, round-trip time and channel values with every digit, and the UTC text is exact.

Frequency setting: SETFSREL

The PLL forces the last IF to equal the synthesizer frequency and the output is the 5 MHz OCXO ([8], Eqs. 10–12, Sec. 9.4).

Table 10. Variables of the T4 frequency setting (F1–F5)

Symbol	Meaning	Unit
F_{out}	hydrogen-maser oscillation frequency in the T4 relations (fixed)	Hz
ν_{ocxo}, ν_0	OCXO frequency; output frequency ($\nu_0 = \nu_{\text{ocxo}} \approx 5$ MHz)	Hz
ν_{synth}	synthesizer frequency (read with `RDFS`)	Hz
$\Delta\nu_{\text{synth}}$, $\Delta\nu_0$	change of synthesizer and of output frequency	Hz
y_0	fractional output change produced by $\Delta\nu_{\text{synth}}$	1
y	measured fractional frequency offset of the T4 (maser minus reference)	1

Table 10 (continued)

Symbol	Meaning	Unit
Δy	output change required, $-y$	1
q_T	synthesizer resolution $5\ \mathrm{MHz}/2^{39} = 9.0949 \times 10^{-6}$	Hz
n_T	integer number of synthesizer steps	steps
$\Delta y_{\mathrm{realized}}$	output change after quantization	1
SETFSREL	entered relative step, mandatory sign, range $\pm 29 \times 10^{-9}$ [9]	s/s
$\mathrm{RDFS}_{\mathrm{pred}}$	predicted RDFS after the entry	Hz

$$\nu_{\mathrm{synth}} = F_{\mathrm{out}} - 284\,\nu_{\mathrm{ocxo}}, \qquad \nu_0 = \nu_{\mathrm{ocxo}} = \frac{F_{\mathrm{out}} - \nu_{\mathrm{synth}}}{284}, \tag{F1}$$

$$y_0 = \frac{\Delta\nu_0}{\nu_0} = \frac{-\Delta\nu_{\mathrm{synth}}}{284 \times 5\ \mathrm{MHz}} = \frac{-\Delta\nu_{\mathrm{synth}}}{1.42 \times 10^9\ \mathrm{Hz}} \qquad \text{(Eq. 12 of [8])}, \tag{F2}$$

$$\Delta y = -y \;\Rightarrow\; \Delta\nu_{\mathrm{synth}} = y \times 1.42 \times 10^9\ \mathrm{Hz}, \qquad q_T = \frac{5\ \mathrm{MHz}}{2^{39}}, \tag{F3}$$

$$n_T = \mathrm{round}\left(\frac{\Delta\nu_{\mathrm{synth}}}{q_T}\right), \qquad \Delta y_{\mathrm{realized}} = -\frac{n_T\, q_T}{1.42 \times 10^9\ \mathrm{Hz}}, \qquad \mathtt{SETFSREL} = \pm\Delta y_{\mathrm{realized}}, \tag{F4}$$

$$\mathrm{RDFS}_{\mathrm{pred}} = \nu_{\mathrm{synth}} + n_T\, q_T. \tag{F5}$$

The resolution q_T is 6.405×10^{-15} relative. [9] defines SETFSREL as a relative step in s/s with a mandatory sign, but not whether $+$ raises the output. The selector "SETFSREL sign" sets the convention, and the panel predicts the new RDFS value (F5), which must lie within 405 708–405 795 Hz. **After entry, compare RDFS with the prediction**; a move in the opposite direction means the convention is reversed. RDFS values near 1.42 GHz (the web page's FREQ) are reduced by 1.42×10^9 Hz.

Drift: DRIFTCOMP

With linear drift the T4 manual models frequency and phase as below ([8], Eqs. 22–23); the quadratic phase fit of Appendix A gives the same quantities.

Table 11. Variables of the T4 drift setting (D1–D3)

Symbol	Meaning	Unit
t	time since the start of the model	s
$y(t)$, y_0	fractional frequency; its value at $t = 0$	1
$\dot{y}$	fractional-frequency drift	s^{-1}
$x(t)$, x_0	phase (time error); its value at $t = 0$	s
c_0, c_1, c_2, d	phase-fit coefficients and elapsed days (Appendix A)	ns, ns/day, ns/day^2; day
r	drift per day, $r = \dot{y} \times 86400$ s from the fit	day^{-1}
ϵ	phase sign, ± 1	1
$\mathtt{DRIFTCOMP}_{\mathrm{now}}$, $\mathtt{DRIFTCOMP}_{\mathrm{new}}$	current (RDREG=DRIFTCOMP;) and new drift compensation	day^{-1}

$$y(t) = y_0 + \dot{y}\,t, \qquad x(t) = x_0 + y_0 t + \tfrac{1}{2}\dot{y}\,t^2, \tag{D1}$$

$$y = \frac{(c_1 + 2c_2 d)\,10^{-9}}{86400}, \qquad r = \frac{2c_2\,10^{-9}}{86400}, \tag{D2}$$

$$\mathtt{DRIFTCOMP}_{\mathrm{new}} = \mathtt{DRIFTCOMP}_{\mathrm{now}} - \epsilon\, r, \qquad 10^{-17} < |\mathtt{DRIFTCOMP}| < 8 \times 10^{-13}. \tag{D3}$$

"A positive sign will generate a positive slope compensation" [9], hence the minus sign in (D3). The entry is followed by REBOOT, which "Reboot[s] the M&C software" ([9], Table 1, p. 8), and RDREG=DRIFTCOMP; to verify. Neither command removes the 1 PPS time offset (ADJ1PPS/SYNCPPS).

Table 12. T4 demonstration with MSM-GPS as a stand-in (not T4 data)

Quantity	Value
Fit: last 30.9 d of 2026 Franks MSM-GPS, quadratic, gap bridged	644 samples
y at MJD 61234.10355	-3.7365×10^{-14}
$\Delta\nu_{\mathrm{synth}}$, steps	-54.570 μHz, -6
Suggested entry ("+ raises output")	`SETFSREL=+3.843E-14;`
RDFS before → predicted after	405750.927598720 → 405750.927544150 Hz
Drift entry (current 0)	`DRIFTCOMP=+7.11e-16;`

Review boundary. No FITS pair or Franks column is identified as the T4 yet. Designate the channel in the GUI ("Selected channel is the T4 comparison") or with `--maser T4 --channel`. T4 settings stay blocked until the channel is designated.

T4 monitoring files (FTP ./Monitoring)

What the manual says

`RECMONIT=<s>;` records the monitoring "in a file stored on the local SD memory card"; the file "can be accessed by FTP under the directory ./Monitoring"; at 2 MB it "is copied with a tag of the date and time, and the current file is reset" ([9], Sec. 1.1.3, p. 12; Sec. 6.5, p. 22). RECMONIT appears only in Sec. 1.1.3, not in the Table 1 command list. The web page shows only whether recording is ON or OFF (Sec. 1.1.3, p. 12; Sec. 4, p. 20). Each record is tagged with the module's date and time, set by NTP or `SETDATE/SETTIME`; without synchronisation at power-up the clock is not kept (pp. 12, 22). Option telemetry such as the HCB is recorded in these files although `MONIT` does not return it (p. 14). The FTP login is in Sec. 6.1 (p. 21). The manual does not document file names, columns or the record layout.

Format, as found in the GBO file

Table 13. Format of the GBO T4 monitoring file `MonitHCB.dat`

Property	`MonitHCB.dat`, GBO T4, received 2026-11-05
Layout	Tab-separated; CR LF; header `TIME` then `Label [unit]` per column
Encoding	UTF-8; the degree sign arrived as U+FFFD (bytes EF BF BD) and is repaired to °
Time	`DD/MM/YY HH:MM:SS`, UTC, 1 s resolution
Records	84: 10 before the time-server sync (3 stamped 2006-01-16, 7 stamped 2007-06-06), then 74 from 17:13:09 to 18:25:39 UTC
Interval	median 60 s, mean 59.59 s, range 59–62 s
Columns	Set, Internal and External temperature [°C]; Alarm Status (text)

Date order. `16/01/06` cannot be MM/DD (no month 16), and YY/MM/DD would date the newest records to 2005. The last record, 05/11/26 18:25:39, was 1.5 min before the file reached this computer (13:27 EST = 18:27 UTC), consistent with UTC as the operator stated. DMY is the default; the reader warns when another order would read every line that the chosen order cannot (a wrong order can still read most lines: `05/11/26` is a valid MM/DD/YY).

Time base and clock-not-set rule

The records carry the maser's own stamps (unlike MONIT rows, which carry this computer's receive time) and may be compared with timing data directly.

Table 14. Variables of the monitoring-file time base and interval statistics (L1–L3)

Symbol	Meaning	Unit
t, t^{Unix}	record time in seconds since 1970-01-01 UTC	s
MJD	Modified Julian Date of the record	day
$t_{\min}$	earliest valid time, "Clock valid from" (default 2020-01-01 UTC)	s
t_{now}	time of reading	s
t_k	successive clock-valid record times	s
Δ_k	interval between successive records	s

$$\mathrm{MJD} = \frac{t^{\mathrm{Unix}}}{86400\ \mathrm{s}} + 40587, \tag{L1}$$

$$\text{clock-valid:}\quad t_{\min} \le t \le t_{\mathrm{now}} + 1\ \mathrm{d}, \tag{L2}$$

$$\Delta_k = t_{k+1} - t_k, \qquad \text{gap if } \Delta_k > 1.5\ \mathrm{median}(\Delta). \tag{L3}$$

So 2026-11-05 17:13:09 UTC is MJD $61349 + \frac{62\,089}{86\,400} = 61349.717465$. Records that are not clock-valid (L2) are listed in the table and CSV (greyed, `clock_valid`=0, with their file and line) but never plotted or analysed ("Clock valid from" in the tab, `--t4-earliest-valid` in batch). Runs of such records are reported separately; a new run starts at a jump of more than 1 h or backwards. Interval statistics use clock-valid records only: the median, mean, minimum and maximum of Δ_k, the count of $\Delta_k < 0$, and the count of gaps (L3).

HCB columns

HCB is the Active Heater-Cooler-Box option (climatic chamber) [13]; the SMC unit sets the temperature of the liquid circulating in its tubes (`SETTEMPHCB`). The HTML page shows "the setting of the temperature", "the internal temperature sensor used for regulation", "the external temperature sensor if used" and "the operation status normal or Alarm (the error code of the error will be displayed)" ([9], Sec. 1.1.3, p. 16). In the GBO file, set point and internal temperature are 21.5 °C throughout the clock-valid span (22 °C in the 2006 pre-sync records), the external temperature is a constant 80 °C, and the status is "No error". A constant 80 °C suggests that no external sensor is connected; this is an inference, not confirmed. Status text is kept as written and plotted as a flag: 0 for "No error", 1 otherwise.

Reading several files

Files with the same header (the live file and its rolled, date-tagged copies) are merged in order of their latest record, so a pre-sync 2006 stamp cannot put the newest file first. Identical records (same time and values) are kept once; records with the same time but different values are kept and reported. Files with different headers stay separate logs. Files are read in parallel threads; at most 2 MB each [9], they need no special memory handling, and "Clear from RAM" releases them.

Outputs and use

- **GUI**: open with File → Open T4 telemetry or monitoring files (Ctrl+T), the T4 menu, or drag and drop; files are routed by content (a tab-separated header starting with `TIME`). The tab shows one x-linked plot per column against MJD − MJD_0, UTC or local date-time, the summary and notes, and every record.
- **Export**: `t4_<layout>_records.csv` (every record, clock flag, source line; UTF-8 with BOM), `t4_telemetry.json`, and `t4_telemetry.vszh5` with one panel per column (each with its own range,

padded by 10 % or ±0.5 when constant), a status page and a notes page, each against MJD − MJD_0 and against local ISO time.

- **Batch**: `analyze_maser.py --t4-telemetry MonitHCB.dat …` adds the pages to `analysis.vszh5` (leave them out with `--no-t4-monitoring-plots`); `python t4_native.py MonitHCB.dat --output NEW_DIR` exports only the monitoring files.

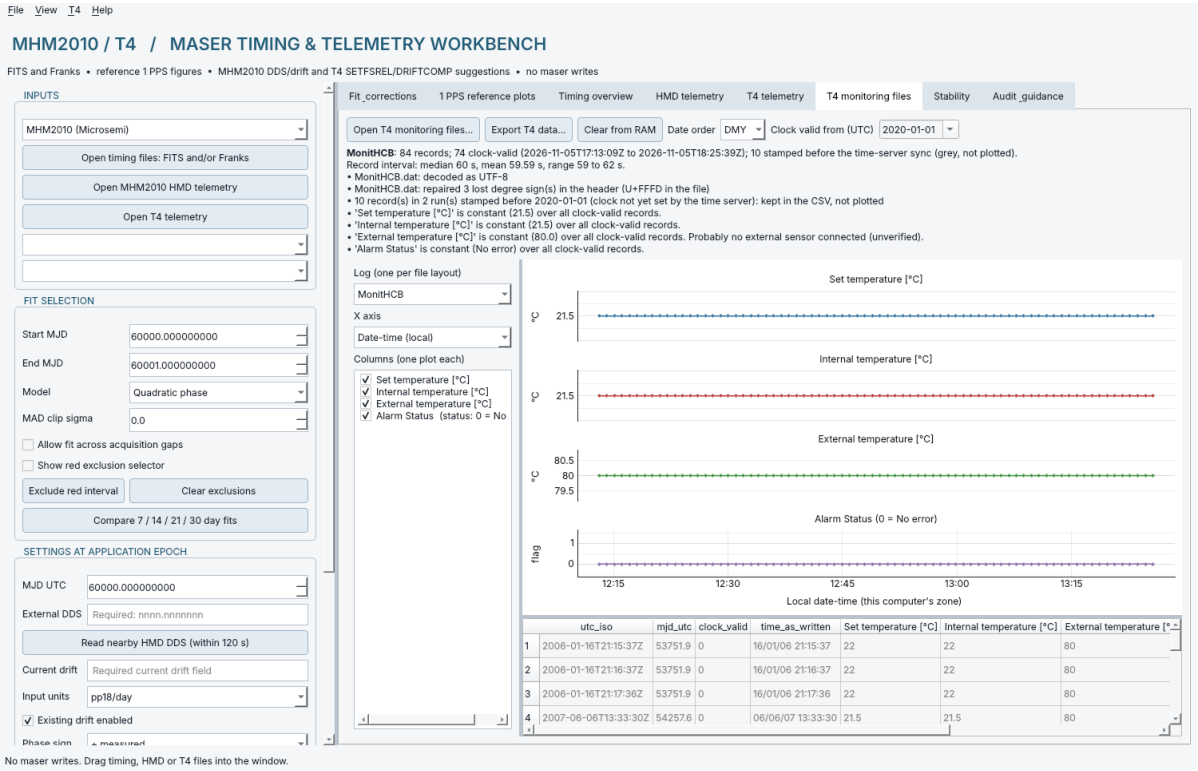


Figure 7. Tab "T4 monitoring files" (Light theme, local date-time axis) on the GBO `MonitHCB.dat`.

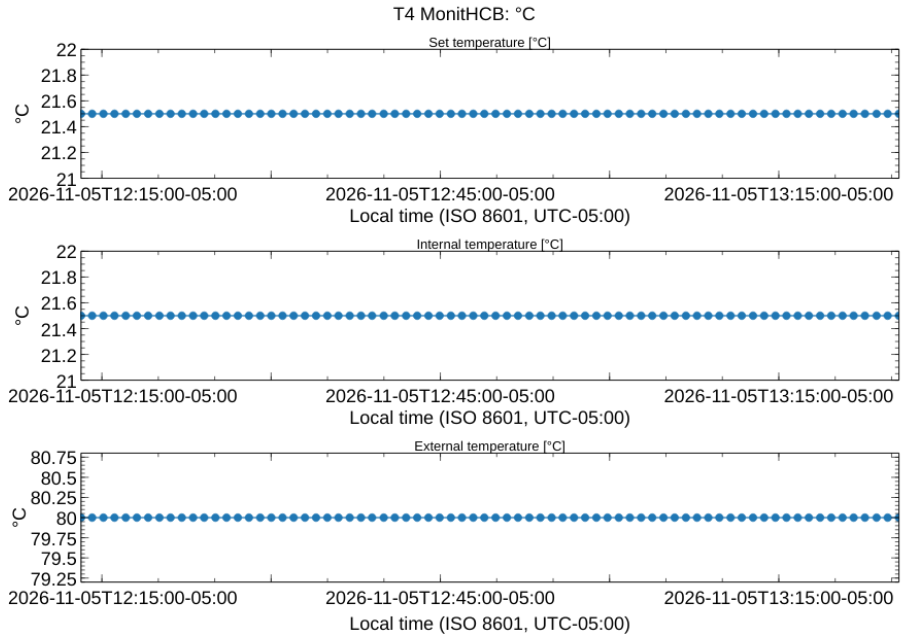


Figure 8. Veusz 3.6.2 render of the exported monitoring page against local ISO time (EST).

Review boundary. The format is inferred from one HCB file. The main 40-channel monitoring file has not been seen; the reader is header-driven and should read it if it has the same layout, but this is untested. HCB alarm codes are not documented here.

Operator script: tools/T4_Reboot.py

The workbench sends no settings and no REBOOT. `tools/T4_Reboot.py` 0.0.4 is a separate operator script that reboots the M&C software after checks; it is untested on a real T4.

1. Read-only checks `GETID;`, `RDFS;`, `RDREG=DRIFTCOMP;` from local port 14000. Each answer must start with `$` and the command; `!<command>` is a rejection ([9], Sec. 1.1.2, p. 11). Any failure stops the script before REBOOT.
2. The operator types `REBOOT`; the command is sent once. Table 1 lists no answer, so silence is the expected result; `!REBOOT` stops the script.
3. After 30 s (the manual's wait after a front-panel reset, Sec. 6.7, p. 24; none is given for REBOOT) the checks are repeated and compared. RDFS may change on its own when drift compensation steps the synthesizer.

Exit codes: 0 success or cancelled, 1 pre-check failed (nothing sent), 2 post-check failed, 3 REBOOT rejected or not sent. The original 0.0.1 script waited for a reply to REBOOT and therefore ended with `TimeoutError` even when the reboot worked (reproduced on loopback). Sec. 1.1.3 (p. 13) describes REBOOT as "Reset the LAN module" and Sec. 6.7 (p. 24) says the equivalent telnet reset "will not affect the stability or the 1PPS generation"; confirm 1 PPS after a reboot.

Stability analysis and Stable32 (base tab)

What the base Stability tab computes

OADEV, MDEV, TDEV and OHDEV use phase samples in seconds and the NIST SP 1065 definitions [2], equations (A25)–(A28) of Appendix A. The calculation supports raw phase, mean removal, linear phase removal and quadratic phase removal. Results are labeled with the chosen detrending; raw stability must not be confused with fitted-residual stability. The traces are described in Appendix C.

The code validates both adjacent intervals and cumulative deviation from a uniform grid, equations (A22)–(A23). It never fills gaps or interpolates samples. The mean actual interval establishes the nominal sampling period after validating the record. The default relative tolerance is 1%; the interface allows up to 5%. This is an explicit approximate-uniformity tolerance, not proof that jitter has no effect at the shortest averaging times.

For the supplied month, a 2% tolerance admits an approximately 3 s maximum grid deviation on a 270 s cadence. The default 1% rejects the full record. An internal exclusion creates a missing interval and is not compressed away to make a false uniform series; such a record is refused here and can be analysed in the gap-aware evaluation tab instead (section "Eval: gap-aware stability").

At least ten contributing terms are required for each plotted estimate. Overlapping terms are correlated; their count is not an equivalent number of independent degrees of freedom. The base tab shows no confidence bands and claims no maser-only noise separation [2]; the evaluation tabs add both (confidence intervals; three-cornered hat).

Which data the stability describes

The stability is computed from the *timing phase*: the measured 1 PPS time difference between the maser and the reference (GPS). It is therefore the stability of the comparison (maser, GPS receiver and link, counter and cables), not of the maser alone. The HMD and T4 telemetry are housekeeping (temperatures, voltages, synthesizer settings); they are not a frequency measurement and cannot give a frequency stability by themselves. Release 0.3.0 uses the HMD only to remove the steering that the maser's own synthesizer applied (section "Eval: steering removal"). A gap-skipping OADEV of the 193-day Franks MSM-GPS record gives 6.3×10^{-13} at 1 h, 6.1×10^{-14} at 1 d and 1.7×10^{-14} at 1 week, while the MHM 2010 datasheet gives 1.5×10^{-15} at 10^4 s and a floor below 1×10^{-15} [24]: the curve measures the GPS comparison out to about two weeks and is only an upper bound on the maser.



Figure 9. Base Stability tab: OADEV, MDEV, OHDEV (upper) and TDEV (lower) of the selected samples.

Stable32 files

`raw.PHD` holds phase seconds and `raw.FRD` holds fractional frequency derived from successive phase samples, equation (A24). The frequency record has one fewer point. Fit-residual versions are written separately. The headers contain type, point count and tau; companion CSV/JSON files retain original timestamps, units, tolerance checks and export decisions [3].

Stable32 treats embedded zeros as gaps. Actual zeros are exported as `1e-99`, following the manual's recommendation; replacements are counted in the sidecar [3]. Do not use its gap-fill controls to disguise a nonuniform or incomplete input record.

Further evaluation (release 0.3.0)

Principles

Four tabs, placed after "Stability" and before "Audit & guidance", evaluate the same data in other ways. Each tab has a banner, a row of controls, a **Calculate** button, plots and a read-only text pane with every number. The **Evaluate** menu has one entry per tab ("Calculate gap-aware stability", "Calculate steering removal", "Calculate Kalman estimate", "Calculate three-cornered hat") and "Tick every evaluation page for export".

- **Evaluation only.** The tabs read the loaded data, the fit interval, the base fit, the application epoch, the phase sign and the maser selection; they write nothing back. The base fit, the Results box, the suggested settings and the reference figures are unchanged (tested). Settings derived from the Kalman estimate are labelled EVALUATION ONLY and appear only in that tab, its pages and its JSON entry.
- **No filled data.** Following NIST SP 1065 [2], a gap stays a gap. For the Allan variance SP 1065 says that a pair touching a gap "is skipped in the summation" (Sec. 10.4); it allows interpolation for analyses that cannot skip, and warns that with many gaps "the results will be suspect" (Sec. 10.6). The workbench skips; it never interpolates. The Kalman filter crosses a gap by prediction alone, so its uncertainty grows there. The grid and gap counts are reported with every result.
- **Shared code.** The GUI tabs and `analyze_maser.py` call the same runners in `maser_eval.py`, so a batch result equals the tab result for the same options.
- **Parallel work.** Averaging factors (and the three pairs of the hat) run in a thread pool; NumPy releases the interpreter lock in the vector operations. The arrays are small (at most about 10^5 samples), so the GPU is not used: the transfer would cost more than the work.

Table 15. The four evaluation tabs: data, method and output

Tab	Data used	Method	Veusz pages / file
Eval: gap-aware stability	selected samples of the base fit, or the steering-removed samples	grid with gap markers; OADEV, MDEV, TDEV, OHDEV skipping gap terms; lag-1 noise type; confidence intervals	`GapStab_*`; `evaluation_gap_stability.csv`
Eval: steering removal	whole selected channel and the HMD DDS column	DDS to fractional frequency, integrated to phase, removed; same fit of the removed record	`Steer_*`; `evaluation_steering_removed.csv`
Eval: Kalman estimate	samples used in the base fit	three-state clock model, Kalman filter and RTS smoother; noise from the gap-aware OHDEV	`Kalman_*`; `evaluation_kalman.csv`
Eval: three-cornered hat	three loaded channels, or two against a common clock	pair OADEVs on common time stamps; Gray–Allan separation	`TCH_allan_deviation`; `evaluation_three_cornered_hat.csv`

Eval: gap-aware stability

Purpose. The base Stability tab refuses a record with gaps or uneven spacing. This tab places the samples on a regular grid, marks empty slots as gaps, computes the four estimators from the terms whose samples all exist, identifies the dominant noise type and gives confidence intervals [2], [18], [17]. On a gap-free uniform record its deviations equal the base tab (tested to 10^{-12} relative) and AllanTools [25].

Table 16. Controls of the tab "Eval: gap-aware stability" and their batch equivalents

Control	Choices (default first)	Batch flag	Meaning
Detrend	Raw phase; Remove mean; Remove linear phase; Remove quadratic phase	`--gap-detrend -1|0|1|2`	polynomial removed before the estimators (G5)
Grid	Auto; Anchored (fixed clock); Sequential (own intervals)	`--gap-grid auto|anchored|sequential`	how samples are placed in slots (G2–G4)
tau0	auto (median interval) or a value in s	`--gap-tau0 S`	grid step τ_0 (G1)
Slot tolerance	10 % of τ_0 (1–50 %)	`--gap-tolerance 0.10`	κ, largest offset placed
Confidence	68.27 % (1 sigma); 90 %; 95 %	`--confidence L`	level L of the intervals (X2, X3)
Data	Selected samples (as the base fit); Steering-removed samples	`--gap-source selected|steering-removed`	the second needs the steering removal

Plots. "Gap-aware frequency stability with confidence intervals" (OADEV, MDEV, OHDEV, log–log, each point with its interval as an error bar) and "Gap-aware time deviation" (TDEV with its interval). The text pane lists the grid (mode, τ_0, samples placed, slots, gap slots, gap runs and the longest, duplicates, largest offset), then one row per estimator and τ: deviation, low and high limits, interval method, EDF, noise type and term count.

Regular grid with gap markers

Table 17. Variables of the regular grid (G1–G5)

Symbol	Meaning	Unit
t_i	time stamp of input sample i, elapsed SI seconds from the grid origin (TAI difference, as (A1))	s
t_o	grid origin: the first valid sample	s
x_i	phase of sample i (maser channel as measured)	s
τ_0	grid step; default the median positive interval	s
κ	slot tolerance, fraction of τ_0, $0 < \kappa \le 0.5$ (default 0.10)	1
k, k_i	slot index; slot of sample i	1
ϕ_i	offset of sample i from its slot time (anchored grid)	τ_0
i'	the previously placed sample (sequential grid)	–
n_i, μ_i	slot advance from i'; its misfit (sequential grid)	slots; τ_0
X_k	grid phase in slot k; empty slot = gap (NaN)	s
N_{in}, N_{pl}, N_{sl}	valid input samples; placed samples; number of slots	1
p_d	detrend order: -1 raw, 0 mean, 1 linear, 2 quadratic	1
$P_{p_d}(t)$	OLS polynomial of order p_d fitted to the placed samples against their real times	s
k_1	first slot holding a sample	1

$$\tau_0 = \text{median}\{t_{i+1} - t_i > 0\} \text{ (unless given)}, \tag{G1}$$

$$\text{anchored:}\quad k_i = \text{round}\,\frac{t_i - t_o}{\tau_0},\quad \phi_i = \frac{t_i - t_o}{\tau_0} - k_i,\quad \text{placed if } |\phi_i| \le \kappa,\quad X_k = x_{\arg\min_{i:k_i=k}|\phi_i|}, \tag{G2}$$

$$\text{sequential:}\quad n_i = \text{round}\,\frac{t_i - t_{i'}}{\tau_0},\quad k_i = k_{i'} + n_i \text{ if } n_i \ge 1 \text{ (else not placed)},\quad \mu_i = \frac{t_i - t_{i'}}{\tau_0} - n_i, \tag{G3}$$

$$\text{auto:}\quad \text{anchored if } N_{\text{pl}}^{\text{anchored}} \ge 0.9\,N_{\text{in}},\ \text{otherwise sequential}, \tag{G4}$$

$$\text{detrend:}\quad X_k \leftarrow X_k - P_{p_d}(t_{i(k)})\ \ (p_d \ge 0),\qquad X_k \leftarrow X_k - X_{k_1}\ \ (p_d = -1). \tag{G5}$$

The anchored grid suits a stable sampling clock (FITS, 270 s); of two samples in one slot the closer is kept and the other counted as a duplicate; samples farther than $\kappa\tau_0$ from a slot are counted as off-grid

and not placed. The sequential grid follows SP 1065 Secs. 5.16 and 10.5 [2] for jittered data such as Franks hourly medians: each sample advances by the rounded number of steps from the previous one, so jitter does not accumulate; an interval of $n_i > 1$ steps leaves $n_i - 1$ gap slots; SP 1065 notes that "tau variations of ± 10 % will yield reasonable results"; intervals with $|\mu_i| > \kappa$ are counted and reported, not rejected. Values are never moved or interpolated; only their slot is chosen. A raw record has a constant removed, which does not change any estimator.

Estimators that skip gaps

Table 18. Variables of the gap-aware estimators (H1–H4)

Symbol	Meaning	Unit
m	averaging factor (integer)	1
τ	averaging time $m\tau_0$	s
Δ_k^2	second difference $X_{k+2m} - 2X_{k+m} + X_k$	s
Δ_k^3	third difference $X_{k+3m} - 3X_{k+2m} + 3X_{k+m} - X_k$	s
S_j	sum of m consecutive second differences, $\sum_{k=j}^{j+m-1} \Delta_k^2$	s
$\mathcal{V}_2$, $\mathcal{V}_3$	slots whose second (third) difference has all its samples	–
$\mathcal{V}_M$	start slots whose m second differences all exist	–
T_2, T_3, T_M	term counts $\lvert\mathcal{V}_2\rvert$, $\lvert\mathcal{V}_3\rvert$, $\lvert\mathcal{V}_M\rvert$ (at least 10)	1
$\sigma_y(\tau)$	overlapping Allan deviation (OADEV)	1
$\operatorname{mod}\sigma_y(\tau)$	modified Allan deviation (MDEV)	1
$\sigma_x(\tau)$	time deviation (TDEV)	s
$\sigma_H(\tau)$	overlapping Hadamard deviation (OHDEV)	1

$$\sigma_y^2(\tau) = \frac{1}{2\tau^2 T_2} \sum_{k \in \mathcal{V}_2} \left(\Delta_k^2\right)^2, \tag{H1}$$

$$\operatorname{mod}\sigma_y^2(\tau) = \frac{1}{2m^2\tau^2 T_M} \sum_{j \in \mathcal{V}_M} S_j^2, \tag{H2}$$

$$\sigma_x(\tau) = \frac{\tau}{\sqrt{3}} \operatorname{mod}\sigma_y(\tau), \tag{H3}$$

$$\sigma_H^2(\tau) = \frac{1}{6\tau^2 T_3} \sum_{k \in \mathcal{V}_3} \left(\Delta_k^3\right)^2. \tag{H4}$$

Without gaps, $T_2 = N_{\mathrm{sl}} - 2m$, $T_M = N_{\mathrm{sl}} - 3m + 1$ and $T_3 = N_{\mathrm{sl}} - 3m$, and (H1)–(H4) are (A25)–(A28). The averaging factors are the base tab's: 35 log-spaced integers from 1 to $\max\left(1, \lfloor (N_{\mathrm{sl}} - 10)/3 \rfloor\right)$, duplicates removed; a point needs at least ten terms. Skipping terms that touch a gap is the SP 1065 rule for the Allan variance [2] and the approach studied by Sesia and Tavella for long periods of missing data [18]; the reported term counts show how much of the record each point uses.

Noise identification (lag-1 autocorrelation)

Table 19. Variables of the lag-1 noise identification (N1–N3)

Symbol	Meaning	Unit
ξ_j	longest gap-free run of the grid, decimated by m, a quadratic removed, then differenced d times	s
$\bar{\xi}$	mean of ξ_j	s
r_1	lag-1 autocorrelation of ξ_j	1
δ	$r_1/(1 + r_1)$	1
d, $d_{\max}$	number of differences; its limit (2 for OADEV, MDEV, TDEV; 3 for OHDEV)	1
$\hat{\alpha}$, α	estimated and rounded power-law exponent, $S_y(f) \propto f^{\alpha}$	1

Table 19 (continued)

Symbol	Meaning	Unit
$S_y(f)$	one-sided spectral density of fractional frequency	Hz^{-1}

$$r_1 = \frac{\sum_j (\xi_j - \bar{\xi})(\xi_{j+1} - \bar{\xi})}{\sum_j (\xi_j - \bar{\xi})^2}, \qquad \delta = \frac{r_1}{1 + r_1}, \tag{N1}$$

$$\text{difference again while } \delta \geq 0.25 \text{ and } d < d_{\max}, \tag{N2}$$

$$\hat{\alpha} = 2 - 2(\delta + d), \qquad \alpha = \mathrm{clip}\big(2 - \mathrm{round}(2\delta) - 2d,\ -2,\ 2\big). \tag{N3}$$

$\alpha = 2, 1, 0, -1, -2$ is white PM, flicker PM, white FM, flicker FM and random-walk FM. The method is that of Riley and Greenhall [17], SP 1065 Secs. 5.5.4–5.5.5 [2]; at least 30 decimated points are needed, otherwise the noise type is "not identified". It is evaluated on the longest gap-free run, so gaps never enter the autocorrelation. The same definitions as AllanTools are used (tested: the same noise type and number of differences).

Equivalent degrees of freedom and confidence intervals

Table 20. Variables of the equivalent degrees of freedom and confidence intervals (X1–X3)

Symbol	Meaning	Unit
N_{eq}	equivalent gap-free number of phase points, $T_2 + 2m$	1
m, α	averaging factor; identified noise exponent (N3)	1
ν_{edf}	equivalent degrees of freedom of the OADEV estimate	1
$\hat{\sigma}$	estimated deviation (any of H1–H4)	1 (s for TDEV)
L	confidence level (default $\mathrm{erf}(1/\sqrt{2}) = 0.6826894921370859$, one standard deviation)	1
β	$1 - L$, the two tails together	1
$\chi^2_p(\nu)$	p-quantile of the chi-squared distribution with ν degrees of freedom	1
z_L	normal quantile $\Phi^{-1}\big((1+L)/2\big)$ (1 for one standard deviation)	1
T	term count of the estimate (T_2, T_M or T_3)	1
σ_{lo}, σ_{hi}	lower and upper confidence limits	as $\hat{\sigma}$

For OADEV, with $N = N_{\mathrm{eq}}$, the simple EDF approximations of SP 1065 Table 5 (Sec. 5.4.1) [2] are used:

$$\nu_{\mathrm{edf}} = \begin{cases} \dfrac{(N+1)(N-2m)}{2(N-m)}, & \alpha = 2, \\[2ex] \exp\sqrt{\ln\dfrac{N-1}{2m}\,\ln\dfrac{(2m+1)(N-1)}{4}}, & \alpha = 1, \\[2ex] \left[\dfrac{3(N-1)}{2m} - \dfrac{2(N-2)}{N}\right]\dfrac{4m^2}{4m^2+5}, & \alpha = 0, \\[2ex] \dfrac{2(N-2)}{2.3N-4.9}\ (m=1), \quad \dfrac{5N^2}{4m(N+3m)}\ (m \geq 2), & \alpha = -1, \\[2ex] \dfrac{N-2}{m(N-3)^2}\left[(N-1)^2 - 3m(N-1) + 4m^2\right], & \alpha = -2. \end{cases} \tag{X1}$$

The chi-squared interval (SP 1065 Sec. 5.3.2, Eq. 45) and, where no EDF applies, the simple interval (Sec. 5.3.1) are:

$$\sigma_{\mathrm{lo}} = \hat{\sigma}\sqrt{\frac{\nu_{\mathrm{edf}}}{\chi^2_{1-\beta/2}(\nu_{\mathrm{edf}})}}, \qquad \sigma_{\mathrm{hi}} = \hat{\sigma}\sqrt{\frac{\nu_{\mathrm{edf}}}{\chi^2_{\beta/2}(\nu_{\mathrm{edf}})}} \qquad \text{(OADEV with a noise type)}, \tag{X2}$$

$$\sigma_{\mathrm{lo}} = \max\left(0,\ \hat{\sigma}\left(1 - \frac{z_L}{\sqrt{T}}\right)\right), \qquad \sigma_{\mathrm{hi}} = \hat{\sigma}\left(1 + \frac{z_L}{\sqrt{T}}\right) \qquad \text{(MDEV, TDEV, OHDEV, or no noise type)}. \tag{X3}$$

The method of each point (`chi2_simple_edf` or `simple_sigma_over_sqrtN`) is written beside it. The simple interval of (X3) uses the term count, which overstates the independent information of overlapping terms, so it is narrower than a full EDF interval would be; it is labelled as such. N_{eq} treats a gapped record as if it were gap-free with the same number of terms, an approximation that SP 1065 does not define for gaps.

Eval: steering removal

Purpose. The MHM2010 output is steered by its external DDS: manual steps and the drift compensation, which changes the DDS slowly (in the historical month `ExtSynth_32` falls by about 13 counts/day). The HMD records the DDS. This tab converts the recorded DDS to the fractional frequency it applied, integrates it to phase and removes it from the measured channel, showing the maser without its recorded steering, then fits the removed record with the base model and window. The constants are those of the base DDS calculation (A13)–(A14) [1].

Table 21. Controls of the tab "Eval: steering removal" and their batch equivalents

Control	Choices (default first)	Batch flag	Meaning
Remove	All DDS changes (steps and drift compensation); Steps only (> threshold)	`--steering-mode all|steps`	which changes are removed (W2)
Step threshold	50 counts	`--steering-threshold 50`	θ: a jump between consecutive HMD rows larger than this is a step
DDS column	`ExtSynth_32`; `Thumbwheels_31`	`--steering-column NAME`	HMD column with the DDS that drives the output
(phase sign)	the side panel's phase sign	`--phase-sign`	ϵ in (W4)
(enable)	Calculate, or the export checkbox	`--remove-steering` (needs `--hmd`)	–

Plots. "Measured and steering-removed phase" (Measured, Steering removed, and Fit of removed record, against time) and "Recorded steering: applied fractional frequency" (from the HMD column). The text pane gives the reference DDS, the HMD rows, the covered samples, the step events, the held HMD intervals and a table of the base fit and the steering-removed fit at the application epoch.

Table 22. Variables of the steering removal (W1–W4)

Symbol	Meaning	Unit
j, T_j	HMD row index; its time (elapsed SI seconds)	1; s
N_j	DDS of HMD row j (column `ExtSynth_32` by default)	DDS units
N_j^{used}	DDS used for row j (all changes, or steps only)	DDS units
N_{ref}	reference DDS: the value used at the first covered timing sample	DDS units
a, q	2×10^{-7}; one count 10^{-7}	1; DDS units
θ	step threshold	counts
$\mathbf{1}[\cdot]$	indicator, 1 when the condition holds	1
Δy_j	output fractional frequency applied by N_j^{used} relative to N_{ref}	1
$\Psi(t)$	applied phase, the integral of Δy with zero-order hold	s
t_i, t_{first}	timing sample time; first timing sample inside the HMD record	s
$x(t_i)$	measured phase of the channel	s
$x^{\mathrm{rem}}(t_i)$	phase with the recorded steering removed	s
ϵ	phase sign, ± 1 (ϵx = maser minus reference)	1

$$\Delta y_j = \frac{D(N_{\text{ref}})}{D(N_j^{\text{used}})} - 1 = \frac{a\,(N_{\text{ref}} - N_j^{\text{used}})}{284.08 + a\,N_j^{\text{used}}}, \qquad D(N) = 284.08 + aN, \tag{W1}$$

$$N_j^{\text{used}} = \begin{cases} N_j & \text{all changes,} \\ N_0 + \sum_{l=1}^{j} (N_l - N_{l-1})\,\mathbf{1}[\,|N_l - N_{l-1}|/q > \theta\,] & \text{steps only,} \end{cases} \tag{W2}$$

$$\Psi(t) = \sum_{l<j} \Delta y_l\,(T_{l+1} - T_l) + \Delta y_j\,(t - T_j), \quad T_j \le t < T_{j+1}; \qquad \Psi(t) \leftarrow \Psi(t) - \Psi(t_{\text{first}}), \tag{W3}$$

$$x^{\text{rem}}(t_i) = x(t_i) - \epsilon\,\Psi(t_i) \qquad (T_0 \le t_i \le T_{\text{last}};\ \text{otherwise not corrected}). \tag{W4}$$

Raising the DDS lowers the output (W1), as in (A14). Between HMD rows the DDS is held (zero-order hold); holds longer than 1 h are listed, because a change inside one would be misplaced in time. Timing samples outside the HMD record are not corrected and are left out of the removed record. The removed record is fitted with the base model order and window (gap bridging as the base), and the frequency and drift at the application epoch are compared with the base fit (S1). The tab notes that the selector codes `IntExtSwitch_33` and `IntExtSoft_34` are not decoded: confirm that the external DDS drives the output. An HMD record that does not overlap the timing data is refused with the two MJD ranges.

Eval: Kalman estimate

Purpose. The base fit assumes one polynomial with independent (white) residuals over the window. The Kalman estimate models the comparison as a clock whose frequency and drift may wander, with the noise levels taken from the data, and gives the frequency and drift at every sample and at the application epoch with uncertainties that reflect the identified noise [19], [20]. It is shown beside the base least-squares values and never replaces them.

Table 23. Controls of the tab "Eval: Kalman estimate" and their batch equivalents

Control	Choices (default first)	Batch flag	Meaning
Noise	Auto: fit to gap-aware OHDEV; Manual values below	`--kalman-noise R Q1 Q2 Q3` (manual)	how R, q_1, q_2, q_3 are set (K7, K8)
R (s^2)	manual default 4×10^{-18}	first value	white PM measurement variance
q1 (s)	manual default 1×10^{-24}	second value	white FM diffusion, $\sigma_y^2 = q_1/\tau$
q2 (1/s)	manual default 1×10^{-32}	third value	random-walk FM diffusion, $\sigma_y^2 = q_2\tau/3$
q3 ($1/s^3$)	manual default 0	fourth value	random-walk drift diffusion
RTS smoothing (plots)	on	`--no-kalman-smoothing` turns it off	smoothed (K6) or filtered states in the plots

Plots. "Kalman fractional frequency ($\pm 1\sigma$) and base fit" and "Kalman drift ($\pm 1\sigma$) and base fit" (the Kalman value with a shaded $\pm 1\sigma$ band and the base least-squares value dashed), and "Innovations (measured minus predicted phase)". The text pane gives the noise parameters and their source, the OHDEV misfit, the normalized innovation squared, the estimate at the application epoch beside the base values and, for an eligible maser/reference pair, the settings from the Kalman estimate marked EVALUATION ONLY.

Table 24. Variables of the Kalman estimate (K1–K9)

Symbol	Meaning	Unit
k, n	sample index; number of samples	1

Table 24 (continued)

Symbol	Meaning	Unit
Δt_k	interval $d_k - d_{k-1}$ between samples (elapsed SI time)	day
$\mathbf{X}_k$	state $(x_k, \dot{x}_k, \ddot{x}_k)^{\mathsf{T}}$: phase, frequency, drift	ns, ns/day, ns/day^2
$\mathbf{F}(\Delta t)$	state transition	–
$\mathbf{Q}(\Delta t)$	process-noise covariance of one step	as $\mathbf{X}\mathbf{X}^{\mathsf{T}}$
q_1, q_2, q_3	diffusion coefficients: white FM, random-walk FM, random-walk drift	s; s^{-1}; s^{-3}
Q_1, Q_2, Q_3	the same in the internal units: $10^{18} \cdot 86400\, q_1$, $10^{18} \cdot 86400^3 q_2$, $10^{18} \cdot 86400^5 q_3$	ns^2/day, ns^2/day^3, ns^2/day^5
R	white PM measurement variance (internally $10^{18} R$ ns^2)	s^2
z_k	measured phase	ns
$\mathbf{H}$	measurement row $(1, 0, 0)$	–
$\hat{\mathbf{X}}_k^-$, $\mathbf{P}_k^-$	predicted state and covariance	–
$\hat{\mathbf{X}}_k$, $\mathbf{P}_k$	filtered state and covariance	–
ν_k, V_k	innovation (measured minus predicted phase); its predicted variance	ns; ns^2
$\mathbf{K}_k$	Kalman gain	–
$\mathbf{C}_k$	smoother gain	–
$\hat{\mathbf{X}}_k^s$, $\mathbf{P}_k^s$	smoothed state and covariance	–
τ, $\hat{\sigma}_H^2(\tau_i)$, $T_{3,i}$	averaging time; gap-aware OHDEV^2 of the raw phase at τ_i; its terms	s; 1; 1
w_i	weight $\sqrt{T_{3,i}}$	1
NIS	normalized innovation squared	1
t_a, t_k	application epoch; last sample at or before it	day
y, r, σ_y, σ_r	frequency and drift at t_a (ϵ applied) and their 1σ	1; day^{-1}
ϵ	phase sign	1

$$\mathbf{X}_k = \mathbf{F}(\Delta t_k)\mathbf{X}_{k-1} + \mathbf{w}_k, \qquad \mathbf{F}(\Delta t) = \begin{pmatrix} 1 & \Delta t & \Delta t^2/2 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{w}_k \sim \mathcal{N}\big(0, \mathbf{Q}(\Delta t_k)\big), \tag{K1}$$

$$\mathbf{Q}(\Delta t) = \begin{pmatrix} Q_1\Delta t + \frac{Q_2\Delta t^3}{3} + \frac{Q_3\Delta t^5}{20} & \frac{Q_2\Delta t^2}{2} + \frac{Q_3\Delta t^4}{8} & \frac{Q_3\Delta t^3}{6} \\ \frac{Q_2\Delta t^2}{2} + \frac{Q_3\Delta t^4}{8} & Q_2\Delta t + \frac{Q_3\Delta t^3}{3} & \frac{Q_3\Delta t^2}{2} \\ \frac{Q_3\Delta t^3}{6} & \frac{Q_3\Delta t^2}{2} & Q_3\Delta t \end{pmatrix}, \tag{K2}$$

$$z_k = \mathbf{H}\mathbf{X}_k + v_k, \qquad v_k \sim \mathcal{N}(0, R), \tag{K3}$$

$$\hat{\mathbf{X}}_k^- = \mathbf{F}\hat{\mathbf{X}}_{k-1}, \qquad \mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^\mathsf{T} + \mathbf{Q}, \tag{K4}$$

$$\nu_k = z_k - \mathbf{H}\hat{\mathbf{X}}_k^-, \quad V_k = \mathbf{H}\mathbf{P}_k^-\mathbf{H}^\mathsf{T} + R, \quad \mathbf{K}_k = \frac{\mathbf{P}_k^-\mathbf{H}^\mathsf{T}}{V_k}, \quad \hat{\mathbf{X}}_k = \hat{\mathbf{X}}_k^- + \mathbf{K}_k\nu_k,$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H})^\mathsf{T} + R\,\mathbf{K}_k\mathbf{K}_k^\mathsf{T}, \tag{K5}$$

$$\mathbf{C}_k = \mathbf{P}_k\mathbf{F}_{k+1}^\mathsf{T}\big(\mathbf{P}_{k+1}^-\big)^+, \qquad \hat{\mathbf{X}}_k^s = \hat{\mathbf{X}}_k + \mathbf{C}_k\big(\hat{\mathbf{X}}_{k+1}^s - \hat{\mathbf{X}}_{k+1}^-\big),$$

$$\mathbf{P}_k^s = \mathbf{P}_k + \mathbf{C}_k\big(\mathbf{P}_{k+1}^s - \mathbf{P}_{k+1}^-\big)\mathbf{C}_k^\mathsf{T}, \tag{K6}$$

$$\sigma_H^2(\tau) = \frac{10}{3}\frac{R}{\tau^2} + \frac{q_1}{\tau} + \frac{q_2\tau}{6} + \frac{11\,q_3\tau^3}{120}, \tag{K7}$$

$$(R, q_1, q_2, q_3) = \arg\min_{\geq 0} \sum_i w_i^2 \left(\frac{\sigma_H^2(\tau_i)}{\hat{\sigma}_H^2(\tau_i)} - 1\right)^2,$$

$$\text{misfit} = \sqrt{\text{mean}_i \left(\frac{\sigma_H^2(\tau_i)}{\hat{\sigma}_H^2(\tau_i)} - 1\right)^2}, \tag{K8}$$

$$\text{NIS} = \frac{1}{n}\sum_k \frac{\nu_k^2}{V_k}; \qquad y = \frac{\epsilon\,\dot{x}(t_a)\,10^{-9}}{86400}, \quad r = \frac{\epsilon\,\ddot{x}(t_a)\,10^{-9}}{86400},$$

$$\sigma_y = \frac{\sqrt{P_{22}}\,10^{-9}}{86400}, \quad \sigma_r = \frac{\sqrt{P_{33}}\,10^{-9}}{86400}. \tag{K9}$$

The model is the three-state clock model of Galleani and Tavella [19] and Zucca and Tavella [20]; (K2) is its exact discretization. The Hadamard variance of the model (K7) is Eq. (23b) of Ishizaki et al. [21] (with q_i there squared) plus the white PM term $(10/3)R/\tau^2$ of a third difference of independent phase errors; the Hadamard variance is not affected by a constant drift, so the fit (K8) is made to the gap-aware OHDEV of the raw phase by non-negative least squares, each point weighted by the square root of its term count. A gap is one longer prediction step: nothing is invented, and the covariance grows across it. The update uses the Joseph form (K5); the prior is diffuse (the first measured phase, zero frequency and drift, variances far larger than the data range). The smoother is that of Rauch, Tung and Striebel [23] (K6). The state at the epoch (K9) is the filtered state at the last sample at or before t_a, predicted forward with (K1)–(K2). NIS (K9) is about 1 when the noise model fits the data; well above 1 means the model is too confident. For settings (evaluation only), the state at the epoch is written as a quadratic about the epoch, $c = (x, \dot{x}, \ddot{x}/2)$ at $d_a = 0$, and passed through the same arithmetic as the base, (S1)–(S4).

Eval: three-cornered hat

Purpose. A comparison measures two clocks together. With three clocks compared in pairs, the individual Allan variances follow if the clocks are uncorrelated [22]. The tab uses the gap-aware OADEV (H1) of each pair on their common time stamps.

Table 25. Controls of the tab "Eval: three-cornered hat" and their batch equivalents

Control	Choices (default first)	Batch flag	Meaning
Mode	Three measured pairs: A-B, B-C, C-A; Two pairs against a common clock C	`--tch-mode three|common`	(Q1)
Channel A-B, B-C, C-A	loaded channels (A-C and B-C in common mode)	`--tch CH1 CH2 [CH3]`	which channel compares which clocks (the operator's statement)
Clock A/B/C name	A, B, C	`--tch-names A B C`	labels of the clocks
Sign	+ measured; - measured (common mode)	`--tch-signs 1 1`	s_1, s_2: bring the channels to A-C and B-C
Detrend each pair	Remove quadratic phase (raw, mean, linear also)	`--tch-detrend 2`	polynomial removed from each pair

Plot. "Three-cornered hat: individual and pair Allan deviations" (Clock A, B, C solid with markers; Pair A-B, B-C, C-A dashed). The text pane gives the mode, the common samples, the grid, the count of non-positive variance estimates per clock, and one row per τ.

Table 26. Variables of the three-cornered hat (Q1–Q3)

Symbol	Meaning	Unit
A, B, C	the three clocks (names chosen by the operator)	–
x_1, x_2, x_3	phase of the chosen channels on the common time stamps	s
s_1, s_2	signs bringing channels 1 and 2 to A-C and B-C (common mode)	1
x_{AB}, x_{BC}, x_{CA}, x_{AC}	pair phases	s
$\sigma^2_{AB}(\tau)$, $\sigma^2_{BC}(\tau)$, $\sigma^2_{CA}(\tau)$	gap-aware Allan variance (H1) of each detrended pair	1
$\sigma^2_A(\tau)$, $\sigma^2_B(\tau)$, $\sigma^2_C(\tau)$	individual Allan variance of each clock	1
$\sigma_A(\tau)$, $\sigma_B(\tau)$, $\sigma_C(\tau)$	individual Allan deviation (only when the variance is positive)	1

$$\begin{aligned} \text{three: } & x_{AB} = x_1,\ x_{BC} = x_2,\ x_{CA} = x_3, \\ \text{common: } & x_{AC} = s_1 x_1,\ x_{BC} = s_2 x_2,\ x_{AB} = s_1 x_1 - s_2 x_2,\ x_{CA} = -s_1 x_1, \qquad \text{(Q1)} \\ & \sigma_A^2 = \tfrac{1}{2}\left(\sigma_{AB}^2 + \sigma_{CA}^2 - \sigma_{BC}^2\right), \qquad \sigma_B^2 = \tfrac{1}{2}\left(\sigma_{AB}^2 + \sigma_{BC}^2 - \sigma_{CA}^2\right), \\ & \sigma_C^2 = \tfrac{1}{2}\left(\sigma_{BC}^2 + \sigma_{CA}^2 - \sigma_{AB}^2\right), \qquad \text{(Q2)} \\ & \sigma_A = \sqrt{\sigma_A^2} \text{ if } \sigma_A^2 > 0, \text{ otherwise not shown (the variance is reported).} \qquad \text{(Q3)} \end{aligned}$$

Common time stamps are those equal to 10^{-7} d (8.64 ms) in every channel; all pairs share one grid (G1–G4) and one detrend. (Q2) neglects the correlation of the clocks, which Gray and Allan show does not bias the result when the clocks are independent [22]; a negative estimate means the assumption or the averaging is not sufficient at that τ. In common mode the pair A-B is derived, so the measurement noise of both channels enters it, and each channel's measurement noise is attributed to its own clock. **Which channel compares which clocks is the operator's statement**; the program cannot know it. If two of the "clocks" share a reference (for example two GPS-derived signals), they are correlated and the separation fails.

Results on the supplied data

The following come from the 0.3.0 export of the historical month (October 15 to November 13, 2025, MicroSem → CNS_GPS1, quadratic, epoch MJD 60992.998201057395, DDS 5751.6772963) with every evaluation ticked, and, for the hat, the 2026 Franks files. Every number is the exported value.

Table 27. Steering removal on the historical month: base fit and steering-removed fit at the application epoch

Quantity (maser minus reference)	Base fit (measured, steered)	Steering removed (maser alone)
fractional frequency y	$5.838444178725387 \times 10^{-15}$	$-2.2198296389965926 \times 10^{-14}$
1σ (conditional OLS)	$2.9491579969249224 \times 10^{-16}$	$2.9508714046127286 \times 10^{-16}$
drift r per day	$-6.60833065550308 \times 10^{-16}$	$-1.5935539121476658 \times 10^{-15}$
1σ (conditional OLS)	$1.9038592212810412 \times 10^{-17}$	$1.9049901361802343 \times 10^{-17}$
samples	9600	9598

The HMD has 43 236 rows (MJD 60963.00147 to 60992.997998), no step above 50 counts and no hold above 1 h; 9598 of the 9810 timing samples are covered. The recorded steering ramps the output by $2.8020162652892274 \times 10^{-14}$ over 29.99062499214051 d (a fitted slope of $9.331958009774105 \times 10^{-16}$/day, $\approx$ 933 pp18/day), and the applied phase totals $3.639978929902729 \times 10^{-8}$ s. **Cross-check:** the drift difference between the two fits is $-9.327208465973579 \times 10^{-16}$/day, the recorded ramp to 0.05 %, and the frequency difference $-2.8036740568691312 \times 10^{-14}$ is the applied $2.8020162652892274 \times 10^{-14}$ to 0.06 %. So in that month the maser itself drifted by about -1.59×10^{-15}/day against GPS; the drift compensation removed about 9.33×10^{-16}/day of it, and the base suggestion acts on the remaining -6.61×10^{-16}/day. The datasheet's $< 2 \times 10^{-16}$/day applies after extended unperturbed operation [24].

Table 28. Gap-aware OADEV of the steering-removed historical month (quadratic removed; anchored grid, $\tau_0 = 270.00026993919164$ s, 9598 slots, no gaps; one-standard-deviation intervals)

τ (s)	OADEV	interval (method)	EDF; noise type
270.00026993919164	$9.869169051129275 \times 10^{-12}$	$9.779278428142905 \times 10^{-12}$ to $9.961584935814033 \times 10^{-12}$ (chi2)	5863.037011054114; flicker PM
3510.0035092094913	$1.0207102741952156 \times 10^{-12}$	$9.996011867934163 \times 10^{-13}$ to $1.0432157340765394 \times 10^{-12}$ (chi2)	1097.2309570122752; white FM
80190.08017193992	$8.157072979077067 \times 10^{-14}$	$7.427862012833443 \times 10^{-14}$ to $9.154018112966076 \times 10^{-14}$ (chi2)	46.46945520966811; white FM
536760.536639113	$1.9294871832440765 \times 10^{-14}$	$1.9037538243232353 \times 10^{-14}$ to $1.9552205421649177 \times 10^{-14}$ (simple)	–; not identified

The record is flicker PM at the shortest averaging time and white FM from about an hour to a day: at those times the comparison is limited by the GPS link and the 1 PPS measurement, not by the maser (datasheet 1.5×10^{-15} at 10^4 s [24]). The interval at the longest τ is the simple one (no noise type with fewer than 30 decimated points) and is too narrow.

Table 29. Kalman estimate on the historical month (fitted samples; automatic noise) beside the base fit

Quantity	Value
noise (fit to gap-aware OHDEV)	$R = 3.4137902027147988 \times 10^{-18}$ s^2, $q_1 = 1.9859909536922152 \times 10^{-22}$ s, $q_2 = 0$, $q_3 = 0$
OHDEV misfit (K8); NIS (K9)	3.3505043228786873; 1.3382282354100896
Kalman y at the epoch	$-3.3827814393397497 \times 10^{-15} \pm 1.7541716157087668 \times 10^{-14}$
Kalman r at the epoch, per day	$-1.212620617688191 \times 10^{-15} \pm 1.013271635171295 \times 10^{-15}$
base y; base r per day	$5.838444178725387 \times 10^{-15} \pm 2.9491579969249224 \times 10^{-16}$; $-6.60833065550308 \times 10^{-16} \pm 1.9038592212810412 \times 10^{-17}$
evaluation-only settings	DDS change $-48.049222131053845 \pm 249.1635452381788$ counts; drift change $+1212.620617688195 \pm 1013.2716351712985$ pp18/day

The noise fit keeps white PM and white FM only. With white FM (a random walk of phase) the frequency at the end of a 30-day window is known to about $\pm 1.75 \times 10^{-14}$, sixty times the conditional OLS value, and the drift to about $\pm 1.0 \times 10^{-15}$/day, larger than the drift change the base suggests. This

agrees with the window sensitivity of section "Historical validation results" (the drift changes sign with the window). The misfit of 3.35 and NIS of 1.34 show that the model does not describe the flicker PM at short τ; the uncertainties are indicative.

Table 30. Three-cornered hat, 2026 Franks files, common mode (MSM-GPS as A-C, CNS_GPS as B-C; clocks MSM, CNS, GPS; quadratic per pair; sequential grid $\tau_0 = 3510.432000271976$ s, 4643 slots, 4333 placed, 310 gap slots in 17 runs)

τ (s)	MSM	CNS	GPS
3510.432000271976	$2.549939126620528 \times 10^{-13}$	$2.5694372621593414 \times 10^{-13}$	$5.793428844083576 \times 10^{-13}$
87760.8000067994	$1.078082531231874 \times 10^{-14}$	$9.87756973943125 \times 10^{-15}$	$5.755705895621374 \times 10^{-14}$
624856.8960484117	$1.7792919617474348 \times 10^{-15}$	$1.101756745065161 \times 10^{-15}$	$1.7960535309336527 \times 10^{-14}$

The GPS term dominates every pair that contains it, as expected. CNS has 7 non-positive variance estimates. Because the Franks values are hourly medians of 1 PPS comparisons, the MSM and CNS values include their channels' measurement noise. Whether CNS is independent of GPS (or GPS-derived) is not known to the program; if it is GPS-derived the separation does not hold.

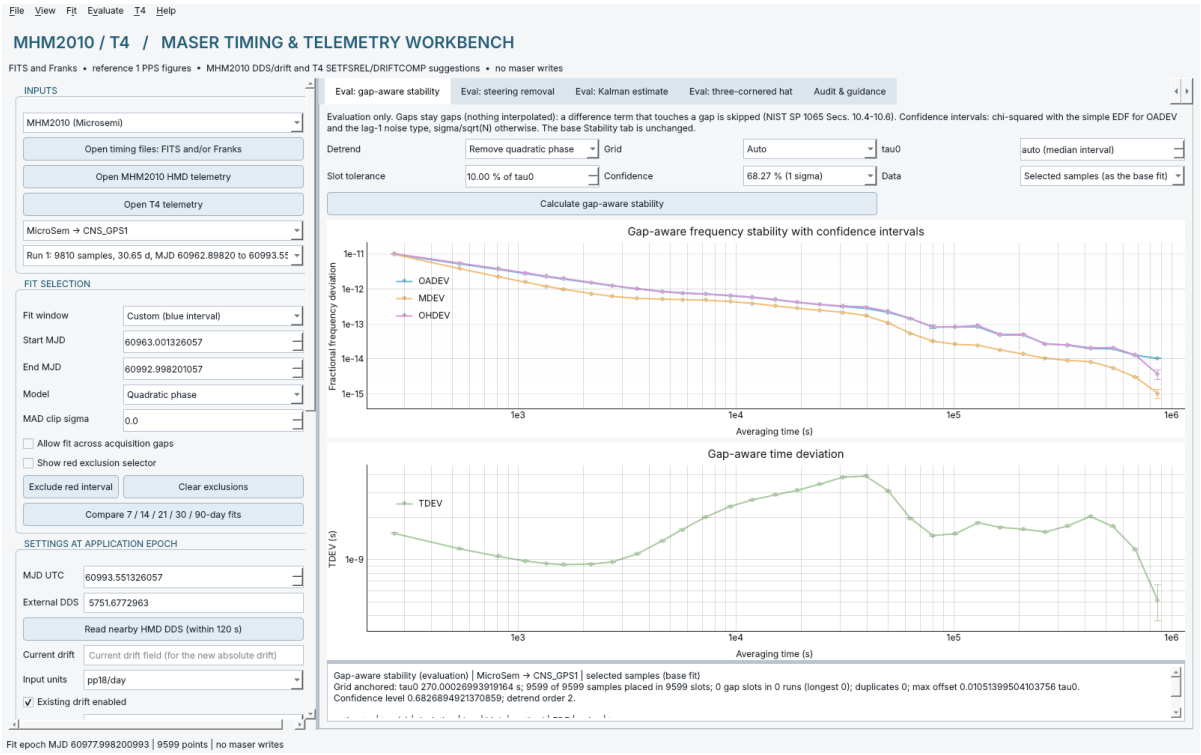


Figure 10. GUI tab "Eval: gap-aware stability" (Light theme): intervals as error bars.

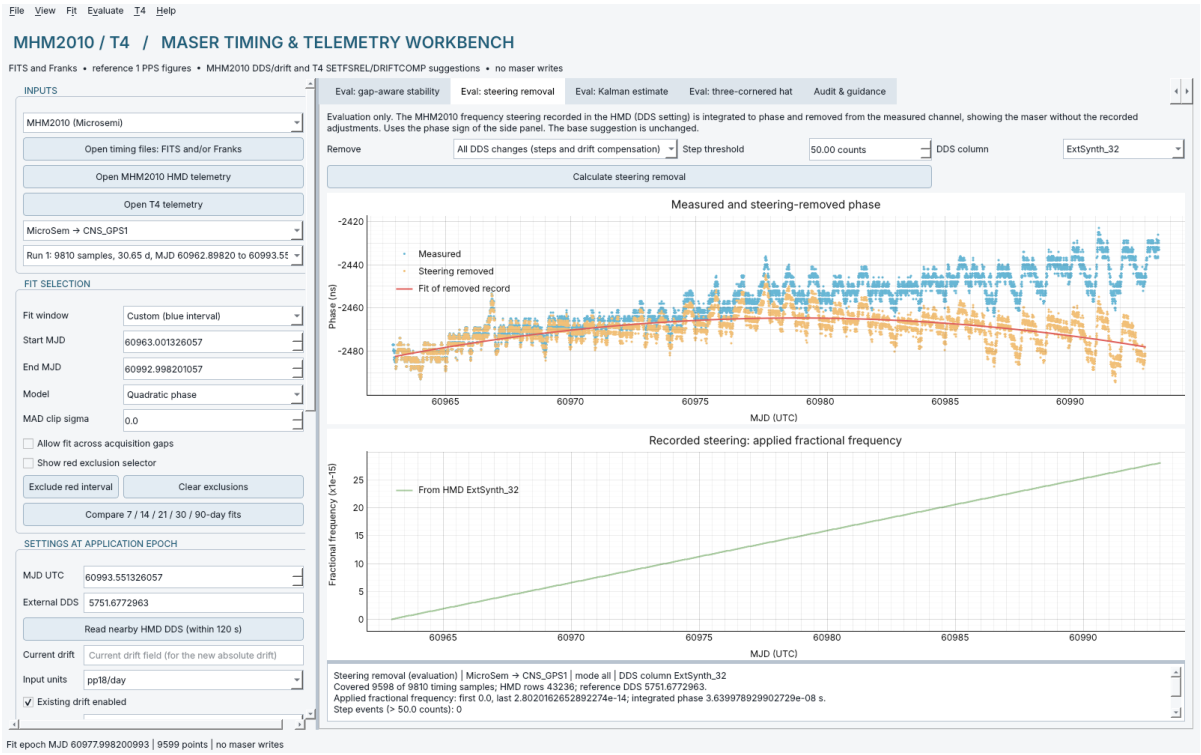


Figure 11. GUI tab "Eval: steering removal" on the historical month.

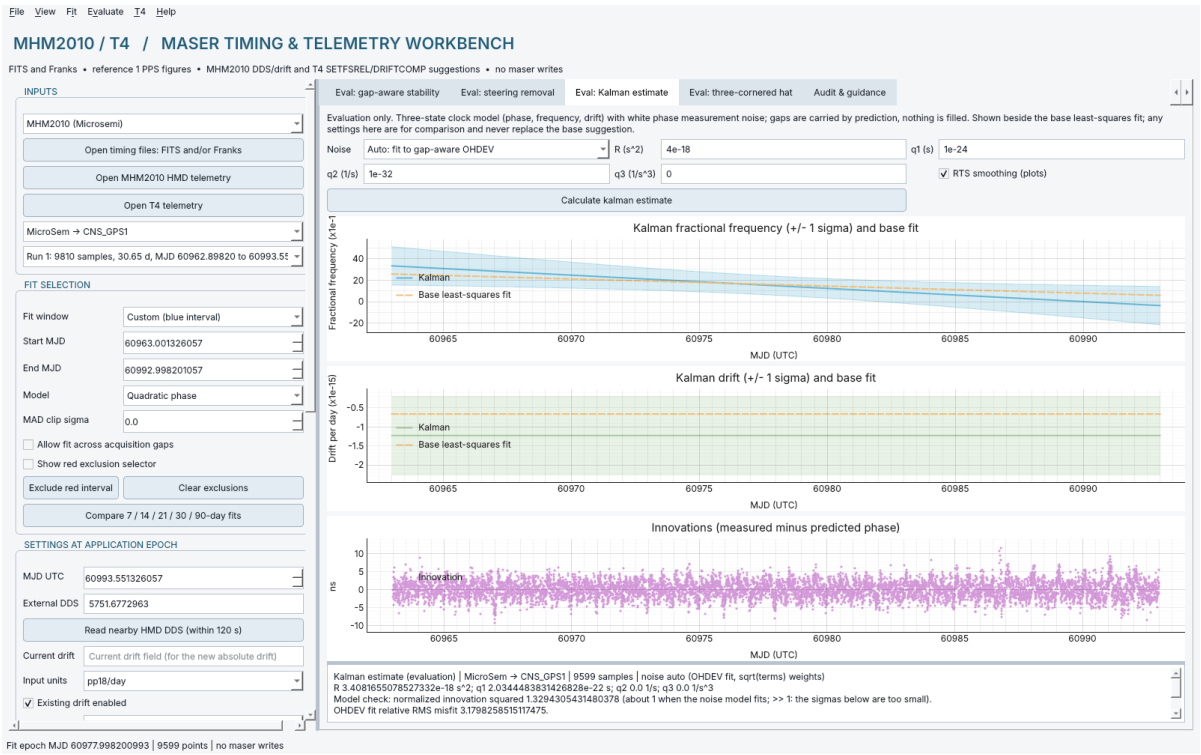


Figure 12. GUI tab "Eval: Kalman estimate" (Light theme): $\pm 1\sigma$ bands, base fit dashed, innovations.

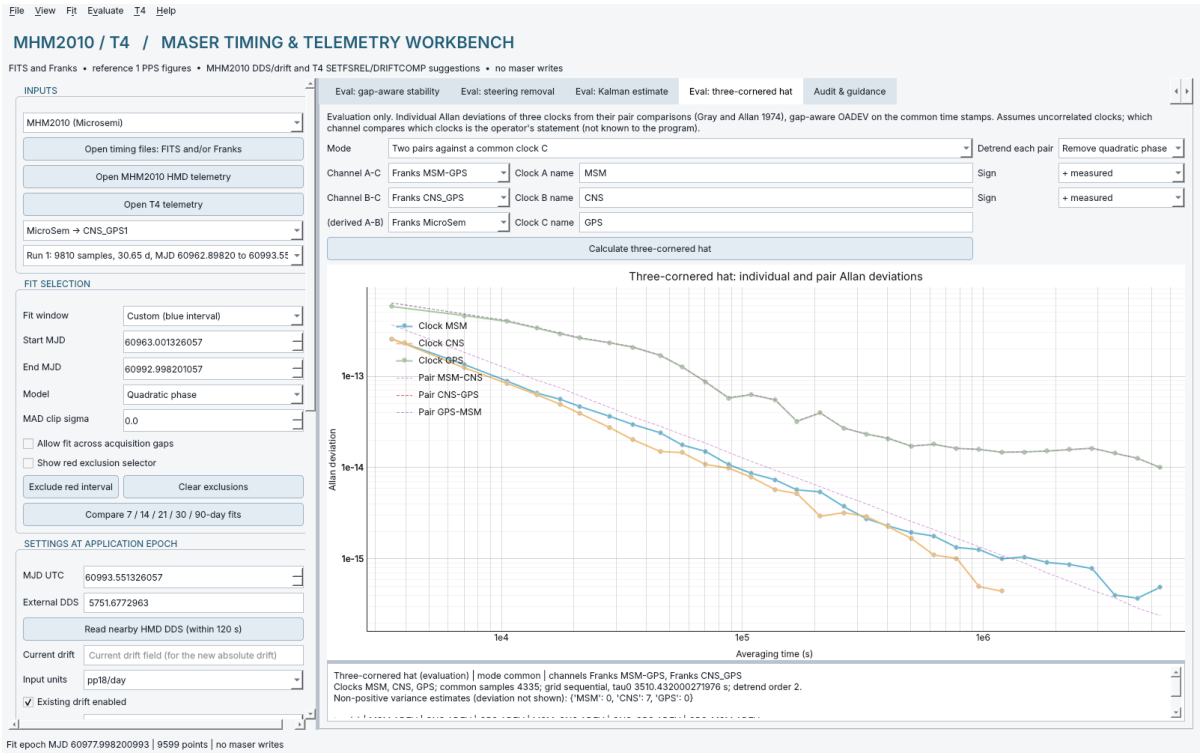


Figure 13. GUI tab "Eval: three-cornered hat" on the 2026 Franks files.

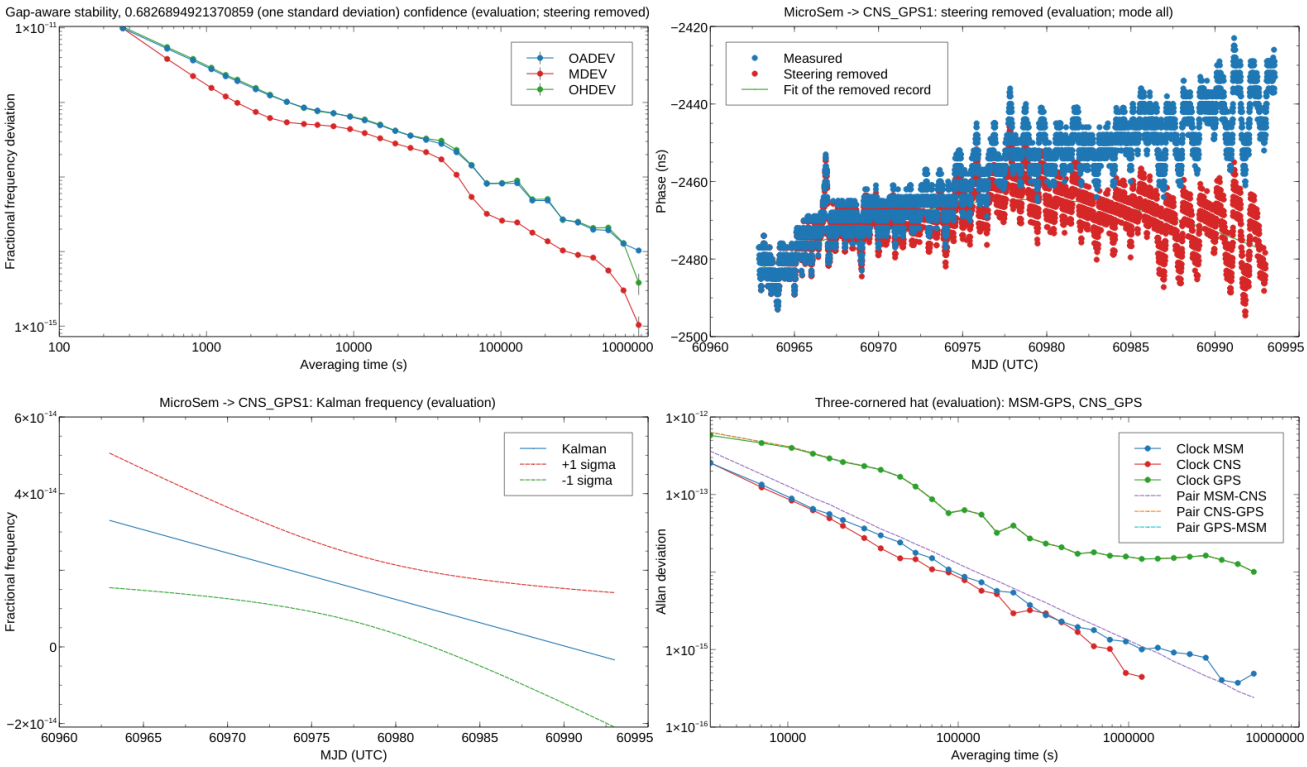


Figure 14. Veusz 3.6.2 renders of the evaluation pages: `GapStab_fractional_frequency`, `Steer_…_phase_mjd`, `Kalman_…_frequency_mjd` and `TCH_allan_deviation`.

Review boundary. The evaluation tabs inform the review; they do not certify a setting. The steering removal depends on the HMD column that drives the output, the Kalman uncertainties on a noise model that fits only approximately, and the three-cornered hat on the operator's statement of which channel compares which clocks and on their independence.

Output package and reproducibility

Files of an analysis export

Table 31. Files of an analysis export folder

Output	Meaning
`analysis.vszh5`	Native HDF5 document, typed datasets and Veusz plotting script; pages as ticked
`analysis.json`	Fit, coefficients, conditional covariance, settings, both drift units, provenance, warnings; 0.3.0: `export_content`, `evaluation`, `evaluation_files` (strict JSON: NaN written as `null`)
`phase_and_fit.csv`	All input timing samples, model, residual, selection and fit masks
`telemetry_overlap.csv`	HMD rows overlapping the selected timing interval, if present
`stability.csv`	Base estimator, tau, deviation and overlapping-term count
`raw.PHD / raw.FRD`	Optional uniform-record Stable32 phase and fractional frequency
`fit_residual.*`	Separately identified residual stability inputs
`evaluation_gap_stability.csv`	0.3.0, when ticked: gap-aware estimators with intervals (Table 32)
`evaluation_steering_removed.csv`	0.3.0, when ticked: measured, applied and steering-removed phase
`evaluation_kalman.csv`	0.3.0, when ticked: Kalman phase, frequency, drift, innovations
`evaluation_three_cornered_hat.csv`	0.3.0, when ticked: clock and pair Allan deviations
`review.md`	Human-readable advisory summary, reference table; 0.3.0 adds "Veusz content" and "Further evaluation"

The HDF5 document is written using the supplied writer without importing Veusz. All measured raw pair datasets and descriptive labels are included when the full collection is exported. Overview medians are separate datasets. The example's externally rendered PDF is a preview, not a dependency of the application.

Exports first build in a temporary adjacent folder and rename only after success. An existing destination is rejected. This protects earlier results on ordinary local filesystems; no distributed-filesystem transactional guarantee is claimed. The CSV, JSON and text files are always complete: unticking a page group removes pages and their datasets from the `.vszh5` only.

Table 32. Columns of the evaluation CSV files (every number exact)

File	Columns
`evaluation_gap_stability.csv`	`estimator`, `tau_s`, `deviation`, `ci_low`, `ci_high`, `ci_method`, `terms`, `edf`, `noise_alpha`, `noise_type`, `differences_d`
`evaluation_steering_removed.csv`	`mjd_utc`, `measured_phase_s`, `applied_fractional_frequency`, `applied_phase_s`, `steering_removed_phase_s`, `covered_by_hmd`
`evaluation_kalman.csv`	`mjd_utc`, `measured_phase_ns`, `phase_ns`, `phase_sigma_ns`, `fractional_frequency`, `fractional_frequency_sigma`, `drift_per_day`, `drift_sigma_per_day`, `innovation_ns`, `innovation_sigma_ns`
`evaluation_three_cornered_hat.csv`	`tau_s`; per clock `clock_<name>_variance`, `clock_<name>_adev`; per pair `pair_<A-B>_adev`, `pair_<A-B>_terms`

Veusz pages to generate

OUTPUT → **Veusz pages to generate** decides which page groups `analysis.vszh5` receives. Base groups are on by default (the file is then the 0.2.4 file); evaluation groups are off. **Defaults** restores that; **All** ticks everything. A page is written only when its data exist (for example, no HMD pages without HMD). A ticked evaluation group is recalculated from the current fit at export time with the options of its tab; if it cannot be calculated (for example steering removal without HMD) it is left out with a note in the audit tab, and the base export continues.

Table 33. Veusz page groups: checkbox, default, batch flag and pages

Checkbox (`ExportContent` name)	Default	Batch flag	Pages (`<ch>` is the channel name)
1 PPS reference figures (with the suggested-changes box) (`reference_figures`)	on	`--no-reference-figures`	`Ref1PPS_<ch>_deltas_MJD`, `_fit_MJD`, `_deltas_local`, `_fit_local`
Selected channel: phase, fit and residual pages (`timing_pages`)	on	`--no-timing-pages`	`Selected_<ch>_phase_and_fit_mjd/_local`, `Fit_<ch>_residual_mjd/_local`
Timing overview of every channel (`overview`)	on	`--no-overview`	`Overview_<channel>_mjd/_local`
Stability (base tab, when calculated) (`stability`)	on	`--no-stability-pages`	`Stability_fractional_frequency`, `Stability_TDEV`
HMD telemetry plots (MHM2010) (`hmd_telemetry`)	on	`--no-hmd-plots`	`HMD_DDS_change_counts_mjd/_local`, `HMD_DDS_setting_mjd/_local`, `HMD_all_channels_MJD_1…`
T4 MONIT telemetry plots (`t4_telemetry`)	on	`--no-t4-plots`	`T4_MONIT_<group>_*`, `T4_MONIT_main_PLL_lock_*`
T4 monitoring-file plots (`t4_monitoring_files`)	on	`--no-t4-monitoring-plots`	`T4log_<layout>_<unit>_*`, `_status_*`
Evaluation: gap-aware stability with confidence intervals (`gap_stability`)	off	`--gap-stability`	`GapStab_fractional_frequency`, `GapStab_TDEV`
Evaluation: steering events removed (needs HMD) (`steering_removed`)	off	`--remove-steering`	`Steer_<ch>_phase_mjd/_local`, `Steer_HMD_<column>_applied_mjd/_local`
Evaluation: Kalman estimate (`kalman`)	off	`--kalman`	`Kalman_<ch>_frequency`, `_drift`, `_innovation` (each `_mjd/_local`)
Evaluation: three-cornered hat (`three_cornered_hat`)	off	`--tch CH …`	`TCH_allan_deviation`

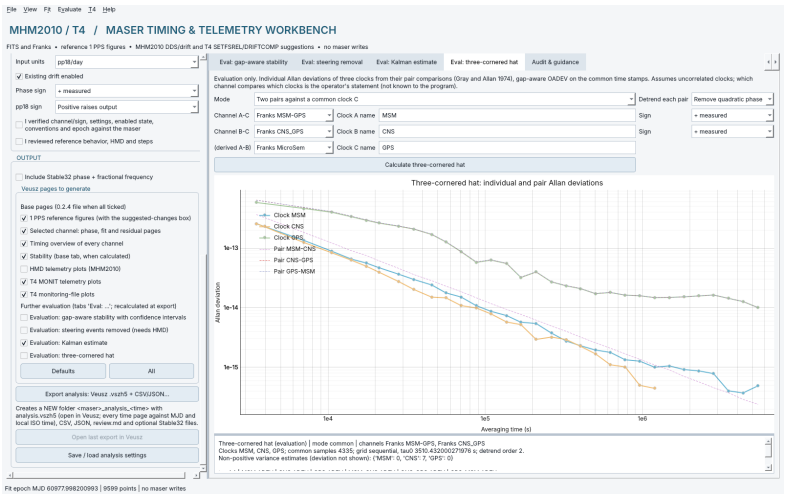


Figure 15. OUTPUT box with the "Veusz pages to generate" checkboxes (Light theme).

The default document of 0.3.0 was compared with 0.2.4 by a fingerprint of every dataset (SHA-256 of the values), every tag, every page and the plotting script (with the creation time removed): identical (`tools/export_fingerprint.py`, `tests/data/export_fingerprint_0.2.4.json`).

Where the Veusz files are

The `.vszh5` files are written only by an export, always into a *new* folder:

Table 34. Where each action writes its Veusz file

Action	Result
File → Export analysis folder (Ctrl+E), or OUTPUT → "Export analysis: Veusz .vszh5 + CSV/JSON"	`<parent>/<MHM2010|T4>_analysis_<YYYYMMDDTHHMMSS±hhmm>/analysis.vszh5` (needs a valid fit)
File → Export T4 data (Ctrl+Shift+E), or the T4 tabs' export buttons	`<parent>/T4_telemetry_export_<time>/t4_telemetry.vszh5`
`analyze_maser.py …--output DIR`	`DIR/analysis.vszh5` (path printed as `veusz_file`)
`t4_native.py FILES --output DIR`	`DIR/t4_telemetry.vszh5`

After a GUI export a dialog shows the full path with **Open in Veusz** and **Open folder**; "Open last export in Veusz" (OUTPUT box and File menu) reopens it. Veusz is looked for at the path set with File → Set Veusz program, then on PATH, then in `Program Files\Veusz`; if not found, the program asks once. Veusz 3.6 or later opens `.vszh5` directly.

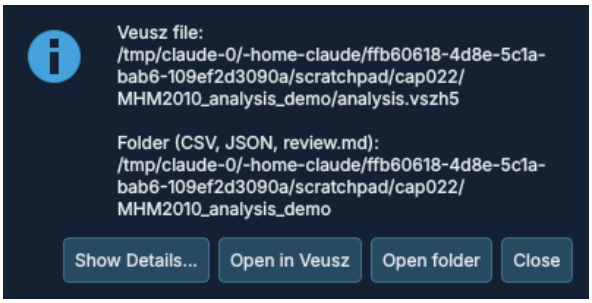


Figure 16. Export dialog with Open in Veusz and Open folder.

Exports: names, tags and full precision

No rounding. Numeric datasets are float64. Numbers written as text use the shortest decimal that reads back to the same binary64 value (Python `repr`, at most 17 significant digits [15]); CSV files are written by pandas without a float format and read back bit-identical (tested). Earlier releases wrote the T4 CSVs with `%.6g` (5751.6770000 became 5751.68) and shortened file hashes in the notes; both are fixed. Figure boxes keep the figure layout; every number in them is also a one-element dataset and a line in `_results_full_precision_text`. Release 0.2.4 extends this to times written as text and to the T4 live recorder (section "Exact times as text"); release 0.3.0 writes the evaluation results the same way (the confidence level in a page title is the full value, `0.6826894921370859 (one standard deviation)`).

Veusz's Data-editor table displays any float with 6 significant digits (Qt's default display; Veusz 3.6.2 `dialogs/dataeditdialog.py`) although the file holds every digit: double-click a cell, or open a `*_text` dataset. Axes use the tick format `%.12Vg` (up to 12 digits; Veusz's automatic `%Vg` shows MJD 60975 as 6.0975×10^4).

Table 35. Dataset names (examples)

Dataset (examples)	Meaning
`Franks_MSM_GPS_phase_ns`, `FITS_Site1Hz_to_CNS_GPS1_phase_ns`	every input channel
`Selected_MSM_GPS_time_MJD_UTC`, `_time_local`, `_phase_ns`	the analysed channel
`Fit_MSM_GPS_model_phase_ns`, `_residual_ns`, `_used_in_fit_flag`	the selected model
`Ref1PPS_MSM_GPS_deltas_ns`, `_fit_line_ns`, `_fit_a1_ns_per_day`	reference figures and their numbers
`Suggested_MHM2010_dds_change_counts`, `_drift_change_pp18_per_day`	suggested changes
`HMD_ExtSynth_32`, `HMD_ExtSynth_32_change_counts`, `_text`	HMD telemetry
`T4log_MonitHCB_Internal_temperature_degC`	T4 monitoring file
`T4_MONIT_ch05_<short>_V`, `_code`	T4 MONIT telemetry
`Stability_OADEV_tau_s`, `_deviation`, `_terms`	base stability
`GapStab_OADEV_deviation` (with ± errors), `_ci_low`, `_ci_high`, `_edf`, `_noise_alpha`	gap-aware stability (0.3.0)
`Steer_<ch>_steering_removed_phase_ns`, `_applied_phase_ns`, `Steer_HMD_ExtSynth_32_DDS_used`	steering removal (0.3.0)
`Kalman_<ch>_fractional_frequency`, `_sigma`, `_plus_1sigma`, `_innovation_ns`	Kalman estimate (0.3.0)
`TCH_tau_s`, `TCH_clock_<name>_adev`, `TCH_pair_<A_B>_adev`	three-cornered hat (0.3.0)

Names use letters, digits and _ only, so each is also a Veusz expression (a pair name such as `MSM-CNS` becomes `MSM_CNS`). With writer 0.0.27 and later the traces take the y dataset name and graphs `g_<y>`, so the widget tree reads like the data. Appendix C lists every stability dataset.

Tags. Every dataset carries one `source_*` tag (`Franks`, `FITS`, `HMD`, `T4_MONIT_UDP`, `T4_log_<layout>`, `Workbench`) and one or more `type_*` tags (`time_MJD`, `time_local`, `time_text`, `measurement`, `fit_model`, `fit_residual`, `fit_results`, `suggested_settings`, `telemetry`, `status`, `stability`, `reference_figure`, …; 0.3.0 adds `evaluation`, `steering`, `kalman` and `metadata`). In Veusz choose Data → *Group by: Tags*; a dataset appears under each of its tags [14]; `type_evaluation` gathers every evaluation dataset.

HMD telemetry. The analysis export holds every HMD column over the fit interval, the DDS columns (`ExtSynth_32`, `Thumbwheels_31`) also as a change in counts from the first value and as exact text, pages for the DDS change and setting, and overview pages of ten panels. In the historical example (Oct.–Nov. 2025) `ExtSynth_32` falls by about 13 counts/day while `Thumbwheels_31` stays at 5751.6774093; release 0.3.0 confirms that this is the drift compensation acting on the DDS: removing it changes the fitted drift by exactly the recorded ramp (section "Results on the supplied data").

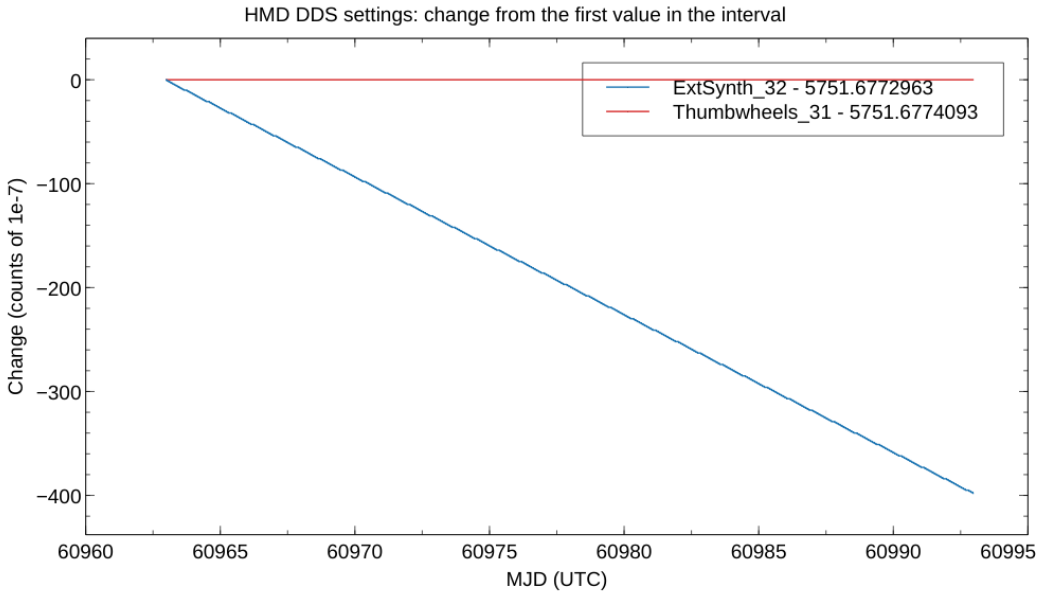


Figure 17. HMD DDS change in counts against MJD (Veusz 3.6.2 render, historical month).

Exact times as text

Releases up to 0.2.3 wrote local times with `datetime.isoformat(timespec="seconds")`, which *cuts* the fraction: the application epoch of the 2026 Franks example, MJD 61234.10355, read `2026-07-12T22:29:06-04:00` although it is 22:29:06.72. Microseconds are not enough either: the spacing of binary64 numbers [15] near the values used here is finer than Python's 1 μs `datetime`:

Table 36. Spacing of binary64 numbers near the time values used

Quantity	Value near	binary64 spacing
MJD (days)	61220	7.28×10^{-12} d $= 0.63$ μs
Unix time (s)	1.79×10^{9}	0.24 μs

Writer 0.0.29 takes the shortest decimal that reads back as the same binary64 MJD (Python `repr`) and converts it in decimal arithmetic [16] with 80 significant digits, so no step rounds.

Table 37. Variables of the exact time text (T1–T2)

Symbol	Meaning	Unit
M	shortest decimal that reads back as the binary64 MJD (Python `repr`)	day
t	Unix seconds of M, in decimal arithmetic	s
$o(t)$	zone offset at instant t (daylight saving included)	s
w	local wall-clock seconds, $t + o(t)$	s
$\lfloor w \rfloor$	whole seconds of w	s
ISO$(\cdot)$, digits$(\cdot)$	ISO 8601 date-time of whole seconds; all decimal digits of the fraction, no trailing zeros	–
text	the written time, e.g. `2026-07-12T22:29:06.72-04:00`	–

$$t = (M - 40587) \times 86400 \text{ s}, \qquad w = t + o(t), \qquad \text{text} = \text{ISO}\big(\lfloor w \rfloor\big) \,.\, \text{digits}\big(w - \lfloor w \rfloor\big) \;\pm \text{HH:MM}, \tag{T1}$$

$$M = \frac{t}{86400} + 40587, \qquad t = w - o. \tag{T2}$$

The inverse (T2), evaluated in decimal arithmetic, gives back the identical binary64 MJD (tested for 10 000 random values in five zones and for every Franks sample) [12]. A Franks MJD has 5 decimals (10^{-5} d $= 0.864$ s), so its text has at most 3 decimals of a second.

Table 38. Time text in 0.2.3 and from 0.2.4 on

Where	0.2.3	0.2.4 and later
`Selected_MSM_GPS_time_local_ISO_text`	`2025-12-31T19:30:06-05:00`	`2025-12-31T19:30:06.624-05:00`
Suggested-changes box	`22:29:06-04:00 (MJD 61234.10355)`	`22:29:06.72-04:00 (MJD 61234.10355)`
MJD − MJD_0 label, FITS M_0	`MJD - 61349.71235`	`MJD - 61349.7123456789`
T4 log text	from the MJD (whole s cut)	the T4's whole seconds, `12:13:09-05:00`
T4 recorder CSV	`%.10f` MJD (8.64 μs), `%.3f`, `%.6g`	every digit (`repr`)

The writer also offers `iso_local_strings(epoch, timespec="exact")` for times that start as Unix seconds (e.g. a host clock); its default (`"seconds"`) is unchanged for every other program. The statistics/fit box keeps the layout of the reference figures; every number in it is a full-precision dataset. A new DDS value is quantized to the 7 decimals the MHM2010 accepts, with the exact change and the residual reported beside it.

MJD and local ISO 8601 time axes

Veusz stores dates as naive seconds from 2009-01-01, so a local wall-clock axis needs the zone offset of each sample.

Table 39. Variables of the Veusz local time axis (V1)

Symbol	Meaning	Unit
V_i	Veusz date value of sample i: naive local seconds since 2009-01-01T00:00:00	s
t_i^{Unix}	Unix seconds of sample i	s
MJD_i	Modified Julian Date of sample i, UTC	day
$o(t_i)$	UTC offset of the processing computer at sample i (daylight saving aware)	s
1 230 768 000	Unix seconds of 2009-01-01T00:00:00 UTC	s

$$V_i = t_i^{\text{Unix}} - 1\,230\,768\,000\ \text{s} + o(t_i), \qquad t_i^{\text{Unix}} = (\text{MJD}_i - 40587) \times 86400\ \text{s}. \tag{V1}$$

The tick format is `%VDY-%VDm-%VDdT%VDH:%VDM:%VDS` followed by the offset, e.g. `2026-11-03T06:31:15-05:00`. Veusz formats are static, so when the data cross a DST change the ticks omit the offset and the axis label lists each offset and when it begins. Each time-series page is written twice (`…_mjd`, `…_local`); the reference figures keep the operator's MJD − MJD_0 axis and gain `_deltas_local` and `_fit_local`. Writer helpers: `mjd_to_epoch`, `local_time_axis`, `iso_local_strings`, `time_pair_pages`.

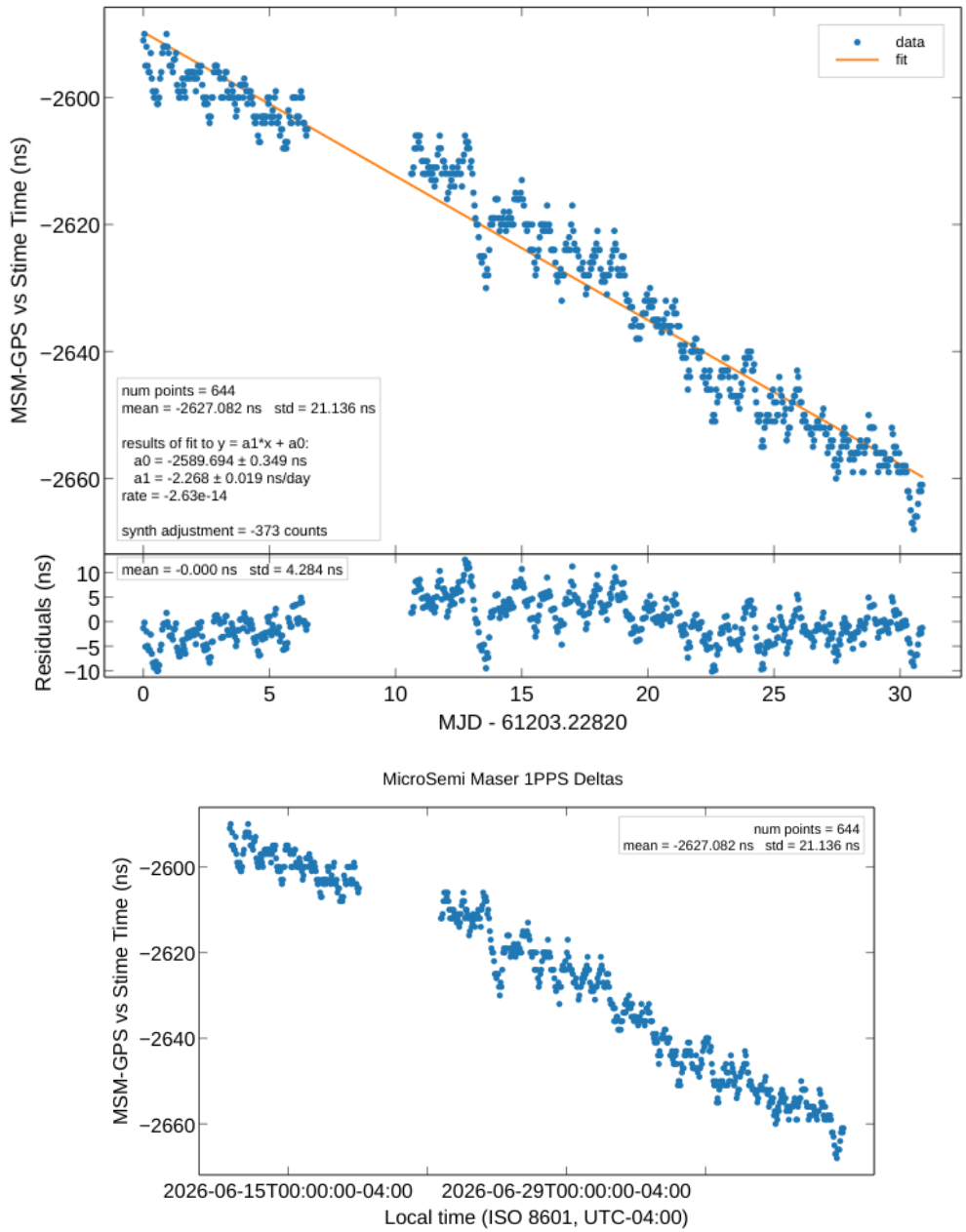


Figure 18. Veusz 3.6.2 renders from `analysis.vszh5`: fit page against MJD − MJD_0, and the deltas twin against local ISO time (EDT in June).

Batch operation

```
python analyze_maser.py --fits-directory <FITS_DIRECTORY>
  --hmd <HMD_FILE> --start 2025-10-15 --end 2025-11-14
  --derive-dds --drift 975 --sampling-tolerance .02
  --stable32 --output <NEW_OUTPUT_DIRECTORY>
```

Join the lines for the shell. The end time is exclusive. Without DDS and drift inputs, the batch path exports measurement results without a correction candidate. `--derive-dds` only copies an HMD setting within 120 s of the historical application epoch; it does not retrieve present-day state.

With the 0.3.0 flags, the following adds every evaluation and leaves the HMD plots out of the `.vszh5` (the HMD CSV and JSON stay complete):

```
python analyze_maser.py --fits-directory <FITS_DIRECTORY> --hmd <HMD_FILE>
  --start 2025-10-15 --end 2025-11-14 --derive-dds --output <NEW_OUTPUT_DIRECTORY>
```

```
  --gap-stability --gap-source steering-removed --gap-detrend 2
  --remove-steering --kalman --no-hmd-plots
python analyze_maser.py --timing-directory <FRANKS_DIRECTORY> --channel MSM-GPS
  --output <NEW_OUTPUT_DIRECTORY> --tch MSM-GPS CNS_GPS --tch-mode common
  --tch-names MSM CNS GPS
```

`--remove-steering` (or `--gap-source steering-removed`) without `--hmd` stops with an error. The printed JSON lists `evaluation_files`. Batch evaluation never changes the printed `suggested_settings`: for the 2026 Franks MSM-GPS channel, last 30 days, the JSON is identical with and without `--gap-stability --kalman --tch`, with a DDS change of -526.7884344769938 counts in both.

Writer revision

The attached writer was version 0.0.21. A targeted test demonstrated that `dataset_date()` accepted multidimensional input, creating invalid date shape or silently flattening it in resizable mode. Version 0.0.22 adds one-dimensional validation before creating the dataset group. Valid callers retain the same API. This is a scoped defect correction, not an exhaustive review of every writer feature.

Release 0.2.0 shipped 0.0.24 (MJD and local ISO time axes). Release 0.2.3 ships 0.0.27, copied unchanged from the operator's FitsAutoPlot folder (SHA-256 `992e46a4…40be2369`); it names graphs and traces after their data. The registry master 0.0.28 differs only in saving speed and writes identical files; see `docs/vszh5_writer_0.0.27_NOTES.md`.

Release 0.2.4 ships 0.0.29: the registry master 0.0.28 (FitsAutoPlot 0.4.1, faster saving, identical files), rebuilt from 0.0.27 and matched to the master byte for byte (SHA-256 `13f68513…f634a64b5`), plus 0.0.29, made from the master for this release (additions only: exact ISO 8601 text, section "Exact times as text"). Every 0.0.28 call writes the same file; see `docs/vszh5_writer_0.0.29_NOTES.md` and `docs/writer_0.0.29/`. Release 0.3.0 ships the same 0.0.29 unchanged (SHA-256 `50631331…4f9d701`); the evaluation pages use its existing helpers (`xy_page`, `time_pair_pages`, `dataset_1d` with error bars).

Historical validation results

The uploaded archive contains 109 FITS.GZ files and 40 measured channel pairs, 9,810 records per pair. The example selects October 15 through November 13, 2025, producing 9,600 direct MicroSem-to-CNS_GPS1 samples. The HMD record covers April 9, 2025, through March 25, 2026. The previously supplied July 2026 timing files do not overlap that HMD record.

At the last selected sample, November 13, 2025, 23:57:24.571 UTC, the quadratic fit gives frequency error $+5.83844 \times 10^{-15}$, residual drift -660.833 parts in 10^{18}/day and phase-residual RMS 5.405 ns. These are computed results from the supplied files, not external instrument specifications.

Review boundary. The example assumes enabled drift $+975$ pp18/day and a positive pp18 sign that raises output frequency. The older screenshot supplies that example number, but does not establish its value in November (the HMD ramp of that month corresponds to about 933 pp18/day, section "Results on the supplied data"). The following candidates are historical and unapproved.

Table 40. Historical example: candidate settings (not for entry)

Quantity	Historical example
HMD DDS near the last timing sample	5751.6772565
Calculated DDS increment, last digits	+82.929597
New rounded DDS	5751.6772648
New drift, pp18/day	+1635.833066
Equivalent new drift, DDS units/day	-23.235467
Nearest integer pp18/day alternative	+1636
Nearest integer DDS units/day alternative	-23

The dominant warning: fit-window sensitivity

Table 41. Historical example: trailing-window sensitivity

Window, d	Frequency error at common end	Drift, pp18/day	RMS, ns
30	$+5.83844 \times 10^{-15}$	-660.833	5.405
21	$+3.80025 \times 10^{-15}$	-883.575	5.810
14	$+1.39421 \times 10^{-14}$	+407.199	5.922
7	-4.32655×10^{-15}	-5454.180	6.970

The drift changes sign with the fitting interval. Conditional OLS errors are much smaller than this variation and must not be interpreted as a complete uncertainty budget. The Kalman estimate of 0.3.0 gives a drift uncertainty of about $\pm 1.0 \times 10^{-15}$/day (1013 pp18/day) for the 30-day window, consistent with this spread. A longer/reference-checked operating-state review is needed before any live adjustment.

For the 30-day window, CNS2018 and CNS_GPS2 comparisons give -688.966 and -652.571 pp18/day. Similar values are useful but do not demonstrate independent references or isolate maser aging.

Cross-checks, testing and limitations

Franks/Veusz comparison

The supplied Veusz example was read as HDF5 data, without executing its embedded plotting script. Across 735 matching UTC-hour bins in the full archive span, the new derived MSM–GPS comparison differs by 1.284 ns RMS from the historical reduced values. MicroSem and CNS_GPS1 differ by 0.852 ns and 1.004 ns RMS. Median differences are approximately zero. The maximum absolute differences are 4, 2 and 4 ns respectively.

These small differences are consistent with nonidentical reduction windows and rounding, but that explanation is not a proof of identical processing. The direct raw pair remains the primary correction input. Invalid negative MJD legacy records and all-missing channels in the exemplar were not copied.

Automated and rendering checks

The 0.1.0 suite had 36 tests; release 0.3.0 has 174 (Python 3.13.18, offscreen Qt). Tests cover analytic phase coefficients, DDS direction, quantization, both drift representations, disabled drift, reversal, duplicates, gap/grid rejection, clipping, FITS metadata/padding, Franks sentinels, GUI controls/settings, writer behavior, export protection, exact exports, names and tags, time text, the T4 and, from 0.3.0, the evaluation methods, page selection and the unchanged suggestion.

OADEV, MDEV, TDEV and OHDEV were compared with AllanTools [25] for seeded synthetic phase data, including contributing-term counts; the gap-aware estimators, noise identification, EDF and intervals were compared with AllanTools too. The 33-page 0.3.0 `analysis.vszh5` with every group ticked was rendered with Veusz 3.6.2 (`validation/veusz_render_v0.3.0/`). GUI captures document the offscreen Qt tests (`validation/gui_0.3.0/`).

Not validated

- **Windows/bracca**: actual desktop operation was not performed (packages are staged when the computer is online).
- **Stable32 executable**: output structure was checked, but proprietary-application import remains a site test.
- **Installed masermon**: sign, range, precision, update cadence, persistence and firmware/version behavior remain site checks.
- **Physical system**: no live adjustment, calibration or post-adjustment monitoring was conducted.
- **Uncertainty**: the 0.3.0 intervals and Kalman uncertainties rest on the identified noise and approximate EDFs; no reference-error removal is claimed beyond the three-cornered hat, whose channel definitions are the operator's statement.
- **HMD selector codes**: which synthesizer drives the output is not decoded (steering removal assumes the external DDS).
- **Veusz 4.2.1**: the new pages were rendered with 3.6.2 only.

Repository provenance

The RF_Equipment_Comms stability-tool code [5] and FitsAutoPlot sources [6] were read for header, GUI and analysis conventions. The science modules in this package are new implementations using the cited definitions, not a wholesale copy of the signal-analyzer tool. Repository checkouts were read-only with respect to their remotes; no push was performed.

Reviewed commits:

- RF_Equipment_Comms: `9305e94e9ad85dbdb94a3765cf0989449dc3f345`.
- FitsAutoPlot: `36add38d973bc07e920ff70680a139241c421f7c`.

The product-page pointer is Microchip MHM-2010 [7].

Validation by release

Release 0.3.0

Table 42. Validation of release 0.3.0

Check	Result
Test suite (Python 3.13.18, NumPy 2.5.4, pandas 3.1.0, SciPy 1.18.1, PySide6 6.12.0, h5py 3.16.0, AllanTools 2024.06; offscreen)	174 passed in 47.08 s: 159 retained (one version check relaxed to "0.2.4 or later"), 15 new
Default export vs 0.2.4	identical fingerprint: every dataset's SHA-256, tags, pages and script
Base suggestion with every evaluation (GUI, batch)	identical suggestion JSON; Franks DDS change -526.7884344769938 counts in both
Gap-free gap-aware estimators	equal to the base tab (10^{-12} relative) and AllanTools (10^{-9})
Gapped OADEV	equal to a brute-force sum over complete terms (10^{-12}); input values untouched
Noise type, EDF, chi-squared limits	noise type and differences identical to AllanTools; EDF and limits to 10^{-12} relative
Sequential grid on jittered Franks data	every deviation inside its interval
Steering removal, synthetic	unsteered phase recovered to 10^{-18} s; a 500-count step found in "steps" mode
Steering removal, historical month	drift difference equals the recorded ramp to 0.05 %
Kalman, simulated clock with a 200-sample gap	frequency within 4σ; NIS between 0.5 and 2; base unchanged
Three-cornered hat, three simulated clocks	each clock within 10 % (three-pair and common modes)
Page checkboxes	unticked groups leave their pages out; CSV/JSON complete
Stale fit after a channel change (0.2.4 defect)	fit cleared; no traceback
Veusz 3.6.2 render, every group ticked	33 pages, 13 of them evaluation pages (`validation/veusz_render_v0.3.0/`)

Not validated: Windows, Veusz 4.2.1 for the new pages, a real T4, the HMD selector codes.

Release 0.2.4

Table 43. Validation of release 0.2.4

Check	Result
Test suite (Python 3.13.18, offscreen)	159 passed: 133 retained, 26 new
Writer 0.0.28 rebuilt from 0.0.27	byte-identical to the registry master (SHA-256 `13f68513…f634a64b5`)
Exact numbers in CSVs	1 628 336 numeric cells checked; none differs from its exact text
Exact time text	10 000 random values in five zones and every Franks sample read back to the same binary64 MJD
Veusz 3.6.2 renders	every page (`validation/veusz_render_v0.2.4/`)
Fresh extract	checksums and all tests pass

Not validated: Windows, Veusz 4.2.1 for these pages, a real T4.

Release 0.2.3

Table 44. Validation of release 0.2.3

Check	Result
Test suite (Python 3.13.18, PySide6 6.12.0, numpy 2.5.4, pandas 3.1.0, offscreen)	133 passed: 112 retained (4 updated for new names) + 21 new
Suggestions vs `correction()` / `t4_settings()`	identical DDS change and new DDS; drift change and new field equal to 10^{-13}, both units, both conventions, both signs
Known rate and drift (synthetic)	recovered within 3σ; signs as expected
Empty DDS and drift fields	both changes shown (0.2.2: none)
Exact exports	CSVs, Stable32, `review.md`, text datasets read back exactly; full SHA-256
Names and tags	every dataset `[A-Za-z][A-Za-z0-9_]*`, tagged, one source tag each
Veusz 3.6.2 renders	`validation/veusz_render_v0.2.3/`

Not validated: Windows, Veusz 4.2.1 for the new pages, a real T4.

Release 0.2.2

Table 45. Validation of release 0.2.2

Check	Result
Test suite (Python 3.12, PySide6 6.12.0, offscreen, TZ America/New_York)	112 passed: 103 from 0.2.1 + 9 time-display, fit-window and export tests
ISO 8601 round trip	`2026-11-05T15:47:13-5:00` → MJD 61349.866123 → `2026-11-05T15:47:13-05:00` / `20:47:13+00:00`; July shows `-04:00`
Every control follows the menu	axes, Start/End/epoch fields, run list, results, status bar, reference, T4 and monitoring tabs
Last 30 / 90 days, 2026 Franks MSM-GPS	622 / 1955 samples; 90 d needs and enables gap bridging; disabled on the 12-day July file
Batch `--start` with `-4:00` offset	runs; `veusz_file` printed; `--last-days 400` refused

Not validated: Windows (time-zone rules come from the operating system there), launching Veusz from the dialog (not installed in the test environment).

Release 0.2.1

Table 46. Validation of release 0.2.1

Check	Result
Test suite (Python 3.12, PySide6 6.12.0, offscreen)	103 passed: 79 from 0.2.0, 9 monitoring-file, 15 reboot-script (loopback)
GBO `MonitHCB.dat`	84 records; 74 clock-valid, 17:13:09–18:25:39 UTC; 10 pre-sync in 2 runs; interval 59–62 s
Date order	DMY reads every line; MDY loses 3 lines and the reader warns
Franks 2026 regression with the file added	unchanged: $N = 644$, $a_1 = -2.268 \pm 0.019$ ns/day, -373 counts
Veusz 3.6.2 render of the monitoring export	5 pages (per-column, status, notes; MJD and local)

Not validated: the main 40-channel monitoring file, the reply port of the real T4, and the reboot script on the real T4.

Release 0.2.0

Table 47. Validation of release 0.2.0

Check	Result
Test suite (Python 3.12.15, PySide6 6.12.0)	79 passed: 36 retained from 0.1.0, 29 T4, 10 reference/Franks/time/export, 4 GUI
Manual example MONIT replies (Type 1 and 2)	Decoded; channel 40 4.765 V vs 4.764 V on the web page
Franks 2026, MSM-GPS, last 30.9 d, linear	$N = 644$, $a_1 = -2.268 \pm 0.019$ ns/day, rate -2.63×10^{-14}, -373 counts (DDS calculation: -372.86)
Writer 0.0.23 master	Rebuilt byte-exact (SHA-256 `90833d76…eaac6`); 0.0.24 adds helpers only
Veusz 3.6.2 renders	25 pages incl. reference, overview and local ISO pages
UDP fault cases (emulator)	timeout, exhaustion, wrong peer, duplicate, garbage, truncation, port in use, cancellation, CR-only replies

Not validated: a real T4 (no GBO captures yet), the SETFSREL sign, Windows, Stable32 import, and Veusz 4.2.1 for the new pages.

Release history

Release 0.2.4

- **Writer up to date.** `vszh5_writer.py` 0.0.29 = the RF-project master 0.0.28 (found in the FitsAutoPlot work; no chat had a newer writer on 2026-11-10) plus 0.0.29 (exact ISO 8601 text). The registry and the master copy are updated.
- **Times as text keep every digit.** The `*_time_local_ISO_text` datasets, the application epoch in the suggested-changes box and the MJD − MJD_0 axis labels. T4 monitoring logs keep the whole seconds the T4 wrote.
- **T4 live recorder.** The CSVs write MJD, round-trip time and channel values with every digit (were `%.10f`, `%.3f`, `%.6g`); the UTC text is exact.
- **Batch.** The printed JSON adds the exact application-epoch MJD.
- **Tests.** 159 pass (133 retained, 26 new).

Release 0.2.3

- **Both suggestions, always.** The MHM2010 has two adjustments, the external DDS and the drift field; the T4 has `SETFSREL` and `DRIFTCOMP`. The workbench states both changes together, at the application epoch, in the Results box, in a box on the reference fit figure (GUI and both Veusz fit pages) and in every export. The changes need no current settings; with the current DDS and drift field entered the new absolute values are added.
- **Exports never round.** Every number written to a CSV, Stable32 file, `review.md`, JSON or Veusz text dataset is the exact float64; every number in a figure box is also a dataset.
- **Names and tags.** Datasets are named `<Source>_<item>_<quantity>[_<unit>]` and tagged `source_*` and `type_*`; traces and graphs carry the same names.
- **Veusz display.** Numeric axes show up to 12 significant digits; the HMD export holds every column; constant series are padded.

Release 0.2.2

- **View → Time display (axes, dates, ranges)**: MJD (UTC), calendar date-time local, or calendar date-time UTC, both ISO 8601 (e.g. `2026-11-05T15:47:13-05:00`). One choice drives every time axis, every date field, the run list and the result text.
- **Fit windows**: "Fit window" presets and a Fit menu with the last 7, 14, 30 and 90 days or all data, each offered only when the loaded data span it; comparison now includes 90 days.
- **Veusz output named**: File → Export analysis folder (Veusz .vszh5, CSV, JSON, Stable32) and Export T4 data (Veusz .vszh5, CSV, JSON); afterwards a dialog gives the `.vszh5` path with Open in Veusz and Open folder.
- MJD tick labels keep the decimals the spacing needs (61349.71, not 61349.7 twice); plot titles are readable in the Light theme.

Release 0.2.1

- **T4 monitoring files**: the LAN module's own logs from FTP `./Monitoring` (e.g. `MonitHCB.dat`) are read by `t4_native.py` 0.0.1, shown in a new tab "T4 monitoring files", exported (CSV, JSON, `.vszh5` against MJD and local time) and added to analysis bundles.

- **Local UDP port 14000** by default for live T4 polling (GUI and command line).
- **Operator script** `tools/T4_Reboot.py` 0.0.4 with loopback tests (it is not part of the workbench, which still sends no settings).
- **Corrections**: REBOOT restarts the M&C software ([9], Table 1, p. 8); manual section and page citations checked page by page.

Release 0.2.0

- **T4 iMaser 3000**: maser selector; SETFSREL and DRIFTCOMP suggestions from the OEM relations [8], [9]; a T4 telemetry tab that reads recorder files and text logs and can poll the maser with read-only queries; a loopback emulator for tests and demonstrations.
- **Reference 1 PPS figures**: the operator's two figures (deltas with statistics; data and fit over residuals) in the GUI and as the first pages of every `analysis.vszh5`.
- **FITS and Franks together**: one selection may hold both; they remain separate series. Franks month-boundary repeats are resolved and reported; fits may bridge gaps.
- **Time axes**: every time-series page exists against MJD (UTC) and against the local date-time of the processing computer, with ISO 8601 tick labels [12] (writer 0.0.24).
- **GUI**: menu bar (File, View with Light/Dark theme, T4, Help).

Release checklist

- **Code**: Python 3.12 syntax, headers, revision history and Spyder cells.
- **GUI**: qtpy/PySide6, pyqtgraph, fitting, overview, HMD, stability, evaluation tabs and audit.
- **Science**: explicit phase sign, epoch, both drift units, no housekeeping-only calibration; evaluation never changes the base suggestion.
- **Outputs**: native HDF5 with selectable pages, raw and reduced datasets, Stable32, evaluation files and provenance.
- **Documentation**: editable LuaLaTeX, references, a variable table before every equation block, Appendices A–C, lists of tables and figures, PDF bookmarks.
- **Limits**: no live writes; Windows, masermon and Stable32 site tests remain.

Release 0.2.0 adds the T4 iMaser 3000 (read-only telemetry and advisory settings); 0.2.1 adds the T4's own monitoring files; 0.2.2 the time display and fit windows; 0.2.3 both suggested changes everywhere and exact, named, tagged exports; 0.2.4 exact time text; 0.3.0 the further evaluation and the page checkboxes. Site tests on the real T4 (captured MONIT replies, RDFS before and after a SETFSREL entry, the main monitoring file) and on Windows remain open; see the validation sections.

Appendix A: Implemented equations

Time, phase and polynomial fit

Elapsed time is SI seconds, obtained by converting UTC timestamps through TAI. The MJD plotting conversion is only a civil-date coordinate; it is not used for elapsed-time derivatives.

Table 48. Variables of the time, phase and polynomial fit (A1–A9)

Symbol	Meaning	Unit
UTC_i, $\mathrm{TAI}(\cdot)$	UTC time stamp of sample i; conversion to TAI	–
t_i, t_0	elapsed SI time of sample i; fit epoch	s
d_i, d	elapsed days from the fit epoch	day
MJD_i	Modified Julian Date of sample i, UTC	day
v_i	Veusz seconds coordinate relative to 2009-01-01 (UTC)	s
h_i	UTC-hour bucket index	1
$\widetilde{\mathrm{MJD}}_h$, $\widetilde{x}_h$	median time and phase of hour bucket h	day; s
x_i	measured phase (time difference) of sample i	s
$\widehat{x}(d)$	fitted phase model	s
c_0, c_1, c_2	phase coefficients	ns; ns/day; ns/day^2
$y(d)$	fractional frequency of the model	1
r	fractional-frequency drift, dy/dd	day^{-1}
s	scaling $\max_i \lvert d_i \rvert$	day
A, A_{ij}	scaled design matrix, $(d_i/s)^j$	1
b_i	phase in ns, $10^9 x_i$	ns
T	scaling matrix $\mathrm{diag}(1, s^{-1}, s^{-2})$, truncated for a linear fit	–
$\widehat{\boldsymbol{\beta}}$, $\mathbf{c}$	scaled and physical coefficient vectors	ns (scaled); as c_j
e_i	fit residual	ns
N, p	number of fitted samples; number of coefficients	1
$\widehat{\sigma}^2$	residual variance	ns^2
C_c	conditional covariance of $\mathbf{c}$	as $c_j c_k$
$\mathrm{RMS}_{\mathrm{ns}}$	residual RMS	ns
$\mathbf{g}$	gradient $(0, 1, 2d)^{\mathsf{T}}$ of $c_1 + 2c_2 d$	1, day
σ_y, σ_r	conditional 1σ of y and of r	1; day^{-1}

$$t_i - t_0 = \mathrm{TAI}(\mathrm{UTC}_i) - \mathrm{TAI}(\mathrm{UTC}_0), \qquad d_i = (t_i - t_0)/86400, \tag{A1}$$

$$v_i = 86400(\mathrm{MJD}_i - 54832), \qquad h_i = \lfloor 24\,\mathrm{MJD}_i \rfloor. \tag{A2}$$

Here v_i is the Veusz seconds coordinate relative to 2009-01-01 and h_i indexes a UTC-hour reduction bucket. Within each nonempty bucket, time and phase are reduced by their medians:

$$(\widetilde{\mathrm{MJD}}_h, \widetilde{x}_h) = (\mathrm{median}_{i:h_i=h}\,\mathrm{MJD}_i,\ \mathrm{median}_{i:h_i=h}\,x_i)\,. \tag{A3}$$

With phase coefficients in nanoseconds and d in days:

$$\widehat{x}(d) = 10^{-9}(c_0 + c_1 d + c_2 d^2)\ \mathrm{s}, \tag{A4}$$

$$y(d) = \frac{10^{-9}}{86400}(c_1 + 2c_2 d), \qquad r \equiv \frac{dy}{dd} = \frac{2c_2\,10^{-9}}{86400}\ \mathrm{day}^{-1}. \tag{A5}$$

For a linear fit, c_2 is not estimated; the drift result is withheld, not presented as evidence of zero physical drift. With the scaled design matrix:

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|A\boldsymbol{\beta} - \mathbf{b}\|_2^2, \qquad \mathbf{c} = T\widehat{\boldsymbol{\beta}}, \tag{A6}$$

$$\widehat{\sigma}^2 = \frac{\sum_i e_i^2}{N - p}, \qquad C_c = T\left[\widehat{\sigma}^2 (A^{\mathsf{T}}A)^{-1}\right] T^{\mathsf{T}}, \qquad e_i = b_i - (A\widehat{\boldsymbol{\beta}})_i. \tag{A7}$$

The reported fit scatter and conditional uncertainties are:

$$\mathrm{RMS}_{\mathrm{ns}} = \sqrt{\frac{1}{N}\sum_i e_i^2}, \qquad \sigma_y = \frac{10^{-9}}{86400}\sqrt{\mathbf{g}^{\mathsf{T}} C_c \mathbf{g}}, \qquad \mathbf{g} = (0, 1, 2d)^{\mathsf{T}}, \tag{A8}$$

$$\sigma_r = \frac{2\,10^{-9}}{86400}\sqrt{(C_c)_{22}}. \tag{A9}$$

They assume IID residuals conditional on the chosen fit; they do not represent a full timing-standard uncertainty (compare the Kalman uncertainties, section "Results on the supplied data").

Clipping, continuity and channel convention

Optional clipping uses median-centered residuals and a robust scale; fitting runs are split at long gaps; the channel sign follows from the pulse arrival times.

Table 49. Variables of clipping, continuity and channel convention (A10–A12)

Symbol	Meaning	Unit
e, e_i	fit residuals	ns
M, S	median of the residuals; robust scale	ns
k	clipping factor (0 disables clipping)	1
ΔMJD_i	interval between successive time stamps	day
$\Delta t_{A\to B}$	measured time difference, channel A to channel B	s
$t_{\text{arrival},A}$, $t_{\text{arrival},B}$	pulse arrival times	s
x_A, x_B	time errors of clocks A and B relative to a common reference	s
$x_{\text{maser}-\text{ref}}$	maser minus reference phase	s
ϵ	phase sign, ± 1	1

$$M = \text{median}(e), \quad S = 1.4826\ \text{median}\,|e - M|, \quad \text{retain } i \text{ if } |e_i - M| \le kS, \tag{A10}$$

$$\Delta\text{MJD}_i > 5\,\text{median}(\Delta\text{MJD})\ \ (\text{run split}), \tag{A11}$$

$$\Delta t_{A\to B} = t_{\text{arrival},B} - t_{\text{arrival},A} = x_A - x_B, \qquad x_{\text{maser}-\text{ref}} = \epsilon\,\Delta t, \quad \epsilon \in \{-1, +1\}. \tag{A12}$$

There are at most eight clipping passes, followed by a final refit. These are software heuristics, not manufacturer operating limits. Fixed cable offsets affect the constant phase term; changing delays can affect slope and drift and are not removed automatically.

DDS correction and both drift units

The manufacturer relationship between the maser frequency, the external synthesizer number and the output is [1]:

Table 50. Variables of the DDS correction and the drift units (A13–A19)

Symbol	Meaning	Unit
f_o, f_m	standard output frequency; maser oscillation frequency	Hz
N, N_{new}	old and new DDS setting	DDS units
a	2×10^{-7}	1
D, D_{new}	denominator $284.08 + aN$; with N_{new}	1
y, y_{new}	measured residual fractional frequency; after the new setting	1
q	least-significant DDS step 10^{-7}	DDS units
k_N, $k_{N,\text{new}}$	a/D; a/D_{new}	DDS units^{-1}
δf_o, δN	small changes of output and DDS	Hz; DDS units
g_{digit}	fractional output change per DDS digit	1
ΔN, Δn	DDS change in DDS units; in digits (counts)	DDS units; counts
round_q	rounding to a multiple of q	–
r	residual drift (A5)	day^{-1}
v_{old}, v_{new}	physical DDS rate	digits/day
P_{old}, P_{new}	displayed drift field in pp18/day	10^{-18}/day
η	pp18 sign: $+1$ positive raises output, -1 positive raises DDS	1

$$f_o = \frac{f_m}{D}, \qquad D = 284.08 + aN, \qquad a = 2 \times 10^{-7}. \tag{A13}$$

For old DDS N, measured residual fractional frequency y, and least-significant DDS step $q = 10^{-7}$:

$$k_N = \frac{a}{D}, \qquad \frac{\delta f_o}{f_o} \simeq -k_N \delta N, \qquad g_{\text{digit}} = -k_N q, \tag{A14}$$

$$\Delta N = \frac{D}{a} y, \qquad N_{\text{new}} = \text{round}_q(N + \Delta N), \qquad \Delta n = \Delta N / q. \tag{A15}$$

Before rounding, the static relation exactly cancels the measured residual:

$$1 + y_{\text{new}} = (1 + y)\frac{D}{D_{\text{new}}}, \qquad D_{\text{new}} = 284.08 + aN_{\text{new}}. \tag{A16}$$

The quantization residual is computed from (A16) with Decimal arithmetic. The physical DDS rate v, in last digits/day, transforms as:

$$v_{\text{new}} = (1 + y)v_{\text{old}} + \frac{r}{k_N q}. \tag{A17}$$

With the displayed pp18/day value P and its sign convention η:

$$v_{\text{old}} = -\frac{\eta P_{\text{old}} 10^{-18}}{k_N q}, \qquad P_{\text{new}} = -\eta\, v_{\text{new}}\, k_{N,\text{new}} q\, 10^{18}, \tag{A18}$$

$$P_{\text{new}} \simeq P_{\text{old}} - \frac{\eta r\, 10^{18}}{1 + y}, \qquad k_{N,\text{new}} = a / D_{\text{new}}. \tag{A19}$$

The code reports the rounded-DDS conversion in the both-units table. An inactive old drift uses $v_{\text{old}} = 0$. These equations describe the local rate at the application epoch, not undocumented firmware scheduling.

Rounding, uniformity and Stable32 conversion

The integer drift alternatives, the uniform-record test of the base stability and the Stable32 conversion use the following.

Table 51. Variables of rounding, uniformity and Stable32 conversion (A20–A24)

Symbol	Meaning	Unit
δr_v, δr_P	drift error of the nearest-integer alternative in DDS counts/day; in pp18/day	day^{-1}
v_{new}, P_{new}, $k_{N,\mathrm{new}}$, q, η	as (A17)–(A19)	–
N	number of samples of the record	1
t_i	elapsed time of sample i (A1)	s
τ_0	mean sample interval	s
ρ	relative interval tolerance, $0 < \rho \le 0.05$ (default 0.01)	1
x_i	phase	s
$\overline{y}_i$	fractional frequency between successive phase samples	1
z, z_{export}	a value and its Stable32 encoding	as the value

$$\delta r_v = -(\mathrm{round}\, v_{\mathrm{new}} - v_{\mathrm{new}}) k_{N,\mathrm{new}} q, \tag{A20}$$

$$\delta r_P = \eta(\mathrm{round}\, P_{\mathrm{new}} - P_{\mathrm{new}}) 10^{-18}. \tag{A21}$$

For a record of N samples the uniformity validation is:

$$\tau_0 = \frac{t_{N-1} - t_0}{N-1}, \qquad \max_i \left| \frac{t_{i+1} - t_i}{\tau_0} - 1 \right| \le \rho, \tag{A22}$$

$$\max_i |t_i - t_0 - i\tau_0| \le \rho\tau_0, \qquad 0 < \rho \le 0.05. \tag{A23}$$

At least eight finite matching samples are required. No missing interval is compressed or interpolated. Stable32 fractional frequency and actual-zero encoding are:

$$\overline{y}_i = \frac{x_{i+1} - x_i}{\tau_0}, \qquad z_{\mathrm{export}} = \begin{cases} 10^{-99}, & z = 0 \text{ (actual observation)}, \\ z, & z \ne 0. \end{cases} \tag{A24}$$

The zero convention follows Stable32's documented gap handling [3].

Stability estimators

All sums are over valid overlapping terms of a validated complete record (A22)–(A23); the gap-aware forms are (H1)–(H4). The implementations follow NIST SP 1065 [2].

Table 52. Variables of the base stability estimators (A25–A28)

Symbol	Meaning	Unit
m	averaging factor, integer ≥ 1	1
τ_0, τ	sample interval; averaging time $m\tau_0$	s
x_i	phase of sample i (detrended as chosen)	s
N	number of samples	1
$\sigma_y^2(\tau)$	overlapping Allan variance (OADEV^2)	1
$\mathrm{mod}\,\sigma_y^2(\tau)$	modified Allan variance (MDEV^2)	1
$\sigma_x(\tau)$	time deviation (TDEV)	s
$\sigma_H^2(\tau)$	overlapping Hadamard variance (OHDEV^2)	1
i, j	summation indices	1

$$\sigma_y^2(\tau) = \frac{\sum_{i=0}^{N-2m-1} (x_{i+2m} - 2x_{i+m} + x_i)^2}{2\tau^2 (N - 2m)}, \tag{A25}$$

$$\operatorname{mod}\sigma_y^2(\tau) = \frac{\displaystyle\sum_{j=0}^{N-3m} \left[\sum_{i=j}^{j+m-1} (x_{i+2m} - 2x_{i+m} + x_i) \right]^2}{2m^2\tau^2 (N - 3m + 1)}, \tag{A26}$$

$$\sigma_x(\tau) = \frac{\tau}{\sqrt{3}} \operatorname{mod}\sigma_y(\tau), \tag{A27}$$

$$\sigma_H^2(\tau) = \frac{\displaystyle\sum_{i=0}^{N-3m-1} (x_{i+3m} - 3x_{i+2m} + 3x_{i+m} - x_i)^2}{6\tau^2 (N - 3m)}. \tag{A28}$$

The displayed OADEV, MDEV and OHDEV are square roots of the respective variances. TDEV is (A27). Averaging factors are log-spaced integers, with duplicates removed and at least ten contributing terms per reported point. Optional detrending subtracts an OLS polynomial of order 0, 1 or 2 before applying these equations; raw mode removes only a constant for numerical convenience.

Appendix B: All variables

Every symbol of the manual, by topic, with its unit and the equations that use it (links). A symbol that means different things in different topics appears once per topic; within a topic it has one meaning.

Table 53. All variables of the manual, by topic

Symbol	Meaning	Unit	Equations
Time, phase and fit (Appendix A)			
UTC_i, $\mathrm{TAI}(\cdot)$	UTC time stamp; conversion to TAI	–	(A1)
t_i, t_0	elapsed SI time; fit epoch	s	(A1), (A22)–(A23)
d_i, d	elapsed days from the fit epoch	day	(A1), (A4)–(A8), (D2)
d_a	elapsed days at the application epoch	day	(S1)
MJD_i	Modified Julian Date, UTC	day	(A2), (A3), (C1), (L1), (V1)
v_i	Veusz seconds coordinate (UTC) from 2009-01-01	s	(A2)
h_i	UTC-hour bucket	1	(A2), (A3)
$\widetilde{\mathrm{MJD}}_h$, $\widetilde{x}_h$	hour-bucket medians	day; s	(A3)
x_i	measured phase	s	(A3), (A24)–(A28)
$\widehat{x}(d)$	phase model	s	(A4)
c_0, c_1, c_2	phase coefficients	ns, ns/day, $\mathrm{ns/day}^2$	(A4), (A5), (S1), (D2)
y, $y(d)$	residual fractional frequency (maser minus reference with ϵ)	1	(A5), (A15)–(A19), (S1)–(S4)
r	fractional-frequency drift	day^{-1}	(A5), (A17), (A19), (S1), (S3), (S4), (D2), (D3)
s	scaling $\max\lvert d_i\rvert$	day	(A6)
A, A_{ij}, T	scaled design matrix; scaling matrix	1; –	(A6), (A7)
b_i, e_i	phase in ns; residual	ns	(A6), (A7), (A8), (A10)
$\widehat{\boldsymbol{\beta}}$, $\mathbf{c}$	scaled and physical coefficients	ns; as c_j	(A6)
N, p	samples in the fit; coefficients	1	(A7), (A8)
$\widehat{\sigma}^2$, C_c	residual variance; coefficient covariance	ns^2; as $c_j c_k$	(A7)–(A9)
$\mathrm{RMS}_{\mathrm{ns}}$, $\mathbf{g}$	residual RMS; gradient	ns; 1, day	(A8)
σ_y, σ_r	conditional 1σ of y, r	1; day^{-1}	(A8), (A9)
M, S, k	residual median, robust scale, clipping factor	ns; ns; 1	(A10)
$\Delta\mathrm{MJD}_i$	time-stamp interval	day	(A11)
$\Delta t_{A\to B}$, t_{arrival}	measured difference; arrival times	s	(A12)
x_A, x_B, $x_{\mathrm{maser-ref}}$	clock time errors; maser minus reference	s	(A12)
ϵ	phase sign ± 1	1	(A12), (S1), (D3), (R4), (W4), (K9)
MHM2010 DDS and drift (Appendix A, suggested settings)			
f_o, f_m	output and maser frequencies	Hz	(A13)
N, N_{new}	DDS setting, old and new	DDS units	(A13)–(A16), (R4)
a	2×10^{-7}	1	(A13)–(A15), (S2), (R4), (W1)
D, D_{new}, $D(N)$	$284.08 + aN$	1	(A13)–(A16), (S2), (W1)
q	one DDS count 10^{-7}	DDS units	(A14)–(A20), (S2), (S4), (R4), (W2)
k_N, $k_{N,\mathrm{new}}$	a/D, a/D_{new}	$\mathrm{DDS\ units}^{-1}$	(A14), (A17)–(A20), (S4)
δf_o, δN, g_{digit}	small output and DDS changes; output per digit	Hz; DDS units; 1	(A14)
ΔN, Δn	DDS change in units; in counts	DDS units; counts	(A15), (S2), (R4)
y_{new}	fractional frequency after the new DDS	1	(A16)

Table 53 (continued)

Symbol	Meaning	Unit	Equations
v_{old}, v_{new}	DDS rate	digits/day	(A17), (A18), (A20), (S4)
P_{old}, P_{new}	drift field in pp18/day	10^{-18}/day	(A18), (A19), (A21)
η	pp18 sign convention	1	(A18), (A19), (A21), (S3)
ΔP, Δv	drift changes in pp18/day and DDS counts/day	10^{-18}/day; counts/day	(S3), (S4)
δr_v, δr_P	drift error of integer alternatives	day^{-1}	(A20), (A21)
Uniform record, Stable32 and base stability (Appendix A)			
N	samples of the record	1	(A22), (A25)–(A28)
τ_0, ρ	mean interval; interval tolerance	s; 1	(A22), (A23)
$\overline{y}_i$, z, z_{export}	Stable32 fractional frequency; a value; its encoding	1; as value	(A24)
m, τ	averaging factor; $m\tau_0$	1; s	(A25)–(A28)
σ_y^2, mod σ_y^2, σ_x, σ_H^2	OADEV^2, MDEV^2, TDEV, OHDEV^2	1; 1; s; 1	(A25)–(A28)
Reference 1 PPS figures (local symbols)			
N_s	samples in the figure	1	(R1), (R3)
x_i, MJD$_0$	MJD$_i$ − MJD$_0$; first sample shown	day	(R2)
u_i, $\bar{u}$, std	1 PPS time difference; mean; population std	ns	(R1), (R2)
$\bar{x}$, S_{xx}, S_{xy}	mean of x_i; sums of squares and products	day; day^2; ns day	(R2), (R3)
a_1, a_0	slope and intercept	ns/day; ns	(R2), (R3)
r_i, σ^2, s	residual; its model variance; its estimate	ns; ns^2; ns	(R3)
$s(a_1)$, $s(a_0)$	standard errors	ns/day; ns	(R3)
y_L	straight-line fractional frequency	1	(R3), (R4)
Time display, exact time text, Veusz axes			
t^{Unix}, t^{local}	Unix seconds; local wall-clock seconds	s	(C1), (V1)
$o(t)$	UTC offset at instant t	s	(C1), (T1), (T2), (V1)
M, t, w	exact decimal MJD; its Unix seconds; local seconds	day; s; s	(T1), (T2)
V_i	Veusz local date value	s	(V1)
T4: MONIT, SETFSREL, DRIFTCOMP, monitoring files			
L_{hex}, c_i, g_i, v_i	frame length; channel code, gain, value	characters; code; V/code; V	(M1)
F_{out}, ν_{ocxo}, ν_0, ν_{synth}	maser, OCXO, output and synthesizer frequencies	Hz	(F1), (F2)
$\Delta\nu_{\text{synth}}$, $\Delta\nu_0$, y_0	changes; fractional output change	Hz; 1	(F2)–(F4)
y, Δy, $\Delta y_{\text{realized}}$	measured offset; required and realized change	1	(F3), (F4)
q_T, n_T	synthesizer resolution; steps	Hz; steps	(F3)–(F5)
`SETFSREL`, RDFS$_{\text{pred}}$	entry; predicted RDFS	s/s; Hz	(F4), (F5)
t, $y(t)$, y_0, $\dot{y}$, $x(t)$, x_0	T4 drift model	s; 1; 1; s^{-1}; s; s	(D1)
`DRIFTCOMP`$_{\text{now}}$, `DRIFTCOMP`$_{\text{new}}$	drift compensation	day^{-1}	(D3)
t_{min}, t_{now}, t_k, Δ_k	clock-valid limits; record times; intervals	s	(L2), (L3)
Gap-aware stability (evaluation)			
t_i, t_o, x_i	sample time and grid origin (elapsed s); phase	s	(G1)–(G5)
τ_0, κ	grid step; slot tolerance	s; 1	(G1)–(G3)
k, k_i, ϕ_i	slot; slot of a sample; its offset	1; 1; τ_0	(G2)
i', n_i, μ_i	previous placed sample; slot advance; misfit	–; slots; τ_0	(G3)
X_k	grid phase (gap = NaN)	s	(G2)–(G5), (H1)–(H4)
N_{in}, N_{pl}, N_{sl}	input, placed samples; slots	1	(G4)
p_d, P_{p_d}, k_1	detrend order; polynomial; first filled slot	1; s; 1	(G5)
m, τ	averaging factor; $m\tau_0$	1; s	(H1)–(H4), (X1)
Δ_k^2, Δ_k^3, S_j	second, third differences; window sum	s	(H1), (H2), (H4)
$\mathcal{V}_2$, $\mathcal{V}_3$, $\mathcal{V}_M$, T_2, T_3, T_M	complete-term sets; term counts	–; 1	(H1)–(H4), (X3)
σ_y, mod σ_y, σ_x, σ_H	OADEV, MDEV, TDEV, OHDEV	1; 1; s; 1	(H1)–(H4)
ξ_j, $\bar{\xi}$, r_1, δ	noise-identification series; mean; lag-1 autocorrelation; $r_1/(1+r_1)$	s; s; 1; 1	(N1)
d, d_{max}	differences taken; limit	1	(N2), (N3)
$\hat{\alpha}$, α, $S_y(f)$	noise exponent; spectral density	1; Hz^{-1}	(N3), (X1)
N_{eq} (written N in X1), ν_{edf}	equivalent points; degrees of freedom	1	(X1), (X2)
$\hat{\sigma}$, σ_{lo}, σ_{hi}	estimate and confidence limits	1 (s for TDEV)	(X2), (X3)

Table 53 (continued)

Symbol	Meaning	Unit	Equations
L, β, z_L, $\chi^2_p(\nu)$, T	level; tails; normal quantile; chi-squared quantile; terms	1	(X2), (X3)
Steering removal (evaluation)			
j, T_j	HMD row; its time	1; s	(W2), (W3)
N_j, N_j^{used}, N_{ref}	recorded, used and reference DDS	DDS units	(W1), (W2)
θ, $\mathbf{1}[\cdot]$	step threshold; indicator	counts; 1	(W2)
Δy_j	applied fractional frequency	1	(W1), (W3)
$\Psi(t)$, t_{first}	applied phase; first covered sample	s	(W3), (W4)
$x(t_i)$, $x^{\text{rem}}(t_i)$	measured and steering-removed phase	s	(W4)
Kalman estimate (evaluation)			
k, n, Δt_k	sample; count; interval	1; 1; day	(K1)–(K9)
$\mathbf{X}_k = (x_k, \dot{x}_k, \ddot{x}_k)^{\mathsf{T}}$	phase, frequency, drift state	ns, ns/day, ns/day^2	(K1), (K9)
$\mathbf{F}$, $\mathbf{Q}$, $\mathbf{w}_k$	transition; process covariance; process noise	–	(K1), (K2), (K4)
q_1, q_2, q_3	diffusion coefficients (SI)	s; s^{-1}; s^{-3}	(K7), (K8)
Q_1, Q_2, Q_3	the same, internal units	ns^2/day^{1,3,5}	(K2)
R, v_k	measurement variance; measurement noise	s^2; ns	(K3), (K5), (K7)
z_k, $\mathbf{H}$	measured phase; measurement row	ns; –	(K3), (K5)
$\hat{\mathbf{X}}_k^-$, $\mathbf{P}_k^-$, $\hat{\mathbf{X}}_k$, $\mathbf{P}_k$	predicted and filtered state and covariance	–	(K4), (K5)
ν_k, V_k, $\mathbf{K}_k$	innovation; its variance; gain	ns; ns^2; –	(K5), (K9)
$\mathbf{C}_k$, $\hat{\mathbf{X}}_k^s$, $\mathbf{P}_k^s$	smoother gain; smoothed state and covariance	–	(K6)
τ, $\hat{\sigma}^2_H(\tau_i)$, w_i	averaging time; measured OHDEV^2; weight	s; 1; 1	(K7), (K8)
NIS, t_a, y, r, σ_y, σ_r	innovation check; epoch; estimate and 1σ	1; day; 1; day^{-1}	(K9)
Three-cornered hat (evaluation)			
A, B, C; x_1, x_2, x_3; s_1, s_2	clocks; channel phases; signs	–; s; 1	(Q1)
x_{AB}, x_{BC}, x_{CA}, x_{AC}	pair phases	s	(Q1)
σ^2_{AB}, σ^2_{BC}, σ^2_{CA}	pair Allan variances (H1)	1	(Q2)
σ^2_A, σ^2_B, σ^2_C; σ_A, σ_B, σ_C	individual Allan variances; deviations	1	(Q2), (Q3)

Appendix C: All stability traces

Every stability trace of the workbench: where it appears, its data and how it is defined. GUI colours are those of the plots (base and gap-aware: OADEV `#67b7d5`, MDEV `#f2c078`, TDEV `#a9cba0`, OHDEV `#d59ada`). "Selected samples" are the samples used in the base fit.

Table 54. All stability traces: GUI, Veusz pages, datasets and file columns

Trace (legend)	Where	Data (datasets, columns)	Definition, interval, notes
Base Stability tab (GUI)			
OADEV	plot "Frequency stability", log–log, circles	selected samples, detrended as chosen; uniform record (A22)–(A23)	(A25); at least 10 terms; no interval
MDEV	plot "Frequency stability"	as OADEV	(A26)
OHDEV	plot "Frequency stability"	as OADEV	(A28); insensitive to a constant drift
TDEV	plot "Time deviation" (s)	as OADEV	(A27)
(note line)	below the controls	τ_0, maximum grid error, detrend	"comparison system, not isolated maser noise"
Base Veusz pages and file (stability ticked; calculated in the tab)			
OADEV, MDEV, OHDEV	page `Stability_fractional_frequency`, title "Comparison-system stability (not isolated maser noise)"	x `Stability_<est>_tau_s`, y `Stability_<est>_deviation`; also `Stability_<est>_terms`	as the GUI traces; tags `source_Workbench`, `type_stability`

Table 54 (continued)

Trace (legend)	Where	Data (datasets, columns)	Definition, interval, notes
TDEV	page `Stability_TDEV`, "TDEV: comparison-system stability"	`Stability_TDEV_tau_s`, `_deviation`, `_terms`	(A27)
(file)	`stability.csv`	`estimator`, `tau_s`, `deviation`, `overlapping_terms`	written whenever calculated, ticked or not
(Stable32 inputs)	`raw.PHD`, `raw.FRD`, `fit_residual.PHD/FRD`	phase; fractional frequency (A24)	for Stable32's own estimators; uniform record only
Eval: gap-aware stability (GUI)			
OADEV	"Gap-aware frequency stability with confidence intervals", circles and error bars	selected or steering-removed samples on the grid (G1)–(G5)	(H1); interval (X2) with the lag-1 noise type, else (X3)
MDEV	same plot	as OADEV	(H2); interval (X3)
OHDEV	same plot	as OADEV	(H4); interval (X3); noise type with $d_{\max} = 3$
TDEV	"Gap-aware time deviation" (s), error bars	as OADEV	(H3); interval (X3)
Gap-aware Veusz pages and file (gap stability ticked)			
OADEV, MDEV, OHDEV	page `GapStab_fractional_frequency`; title gives the confidence level (full value) and the data source	x `GapStab_<est>_tau_s`, y `GapStab_<est>_deviation` with + error $\sigma_{\text{hi}} - \hat{\sigma}$ and − error $\sigma_{\text{lo}} - \hat{\sigma}$; also `_ci_low`, `_ci_high`, `_terms`, `_edf`, `_noise_alpha`	tags `type_evaluation`, `type_stability`; grid and method in `GapStab_grid_and_method_JSON`
TDEV	page `GapStab_TDEV`	`GapStab_TDEV_*` as above	(H3)
(file)	`evaluation_gap_stability.csv`	`estimator`, `tau_s`, `deviation`, `ci_low`, `ci_high`, `ci_method`, `terms`, `edf`, `noise_alpha`, `noise_type`, `differences_d`	one row per estimator and τ
Kalman noise model (not plotted)			
measured and model OHDEV	`Kalman_<ch>_summary_JSON`, `analysis.json` (`evaluation.kalman.noise.detail`)	`tau_s`, `ohdev`, `model_ohdev`, `rms_relative_misfit`	gap-aware OHDEV of the raw phase (H4) and the model (K7), fit (K8)
Eval: three-cornered hat (GUI)			
Clock A, B, C	"Three-cornered hat: individual and pair Allan deviations", solid with markers (`#67b7d5`, `#f2c078`, `#a9cba0`)	pair OADEVs on common time stamps	(Q2), (Q3); a non-positive variance leaves a gap in the trace
Pair A-B, B-C, C-A	same plot, dashed (`#d59ada`, `#e05f5b`, `#9fa8da`)	detrended pair phases (Q1)	(H1)
Three-cornered hat Veusz page and file (hat ticked)			
Clock `<name>`	page `TCH_allan_deviation`	x `TCH_tau_s`, y `TCH_clock_<name>_adev`; also `TCH_clock_<name>_variance` (negative values kept)	(Q2), (Q3); method in `TCH_method_JSON`
Pair `<A-B>`	same page, dashed	`TCH_pair_<A_B>_adev`	(H1)
(file)	`evaluation_three_cornered_hat.csv`	`tau_s`, `clock_<name>_variance`, `clock_<name>_adev`, `pair_<A-B>_adev`, `pair_<A-B>_terms`	one row per τ

The reference figures' "std" box value (R1) is a scatter, not a stability estimate; the Kalman frequency and drift traces are time series, not stability traces (section "Eval: Kalman estimate").

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